RB4107 ES1 rescue carrier, 1:5 prototype

Track segments: complete pack, printed-track version (Nazim, 16 Oct 2026)

Two versions of this pack exist: this one with my printed track, and its companion with Nhien's bought belt (B10). Frame, wheels, shafts, bearings and every joint are identical; only the track pages, the procurement list and the deck note differ.

Interactive 3D assembly (slide it apart and back together in 11 steps, bend the hinge, switch the track, click any part):

https://claude.ai/artifact/6gnfWidMkD9kDk2BQc1tBw

PART A: my printed track on Nhien's Rev B frame (his side plates P1 / P2 and the parts in his BOM)

1. How it works: assembled with the 15 numbered joints and the drive train (16, 17); bent down the ES1 stairs; where the hinge stops
2. Piece it together: five of the 11 build steps, connection by connection
3. Parts catalogue, one page per part, highlighted in the assembly, with clickable vendor links
4. Procurement list: track side and Nhien's, links, prices, delivery (clickable)
5. Layout with Nhien's plates: where everything sits, what changed
6. Connection map: one side pulled apart, every joint with its fastener
7. Drawing TS-01 Track (the part I print) and TS-06 Toothed wheel (bought plain-bore, bored at the workshop), each with its explainer

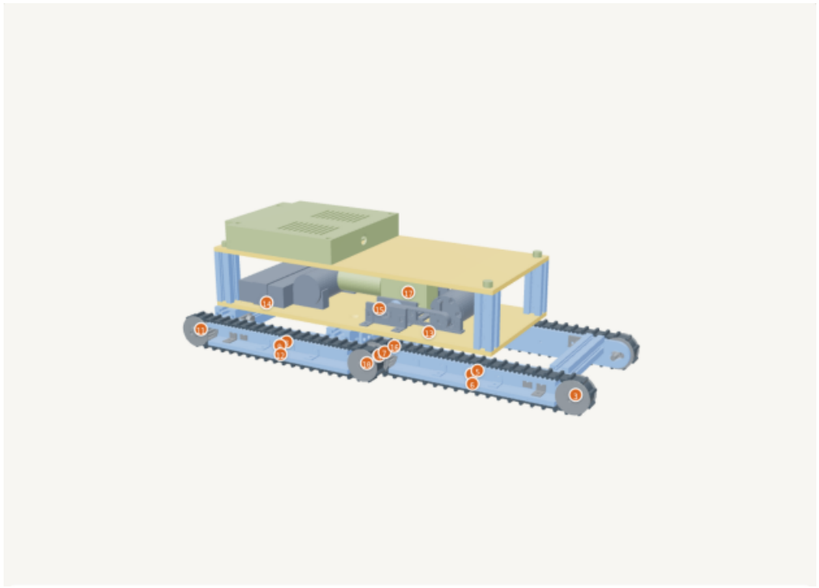
PART B: Nhien's Rev B sheets for the parts the track runs on, unchanged from his fabrication pack

8. Assembly views and joint schedule; P1 and P2 side plates; P3 to P6 rails; P7 / P8 bars; P18 / P19 brackets; P17 support; P20 shafts; M1 to M8 modifications; P11 sector gear and P14 angle; his BOM

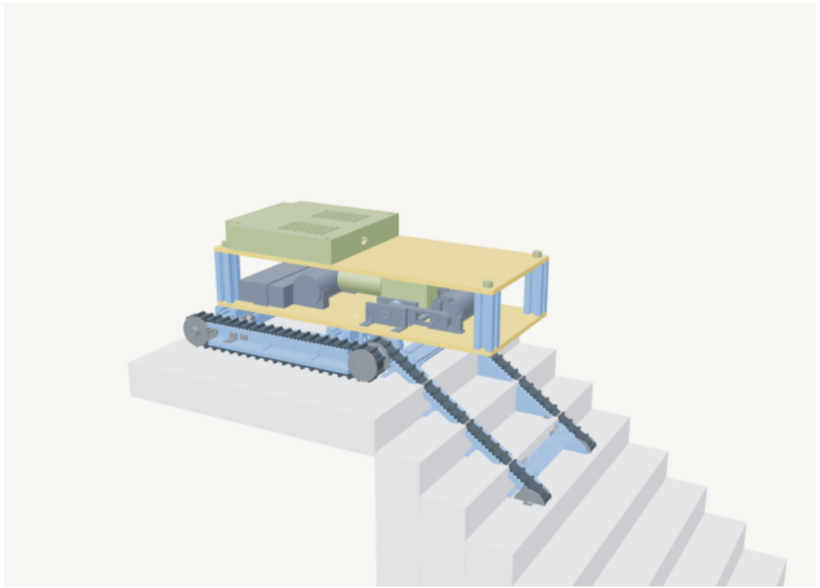
Files on the branch: track_print/stl (print files), track_print/step (CAD, STEP), track_print/viewer (this interactive page), track_print/assembly (both packs and their sources).

1. How it works

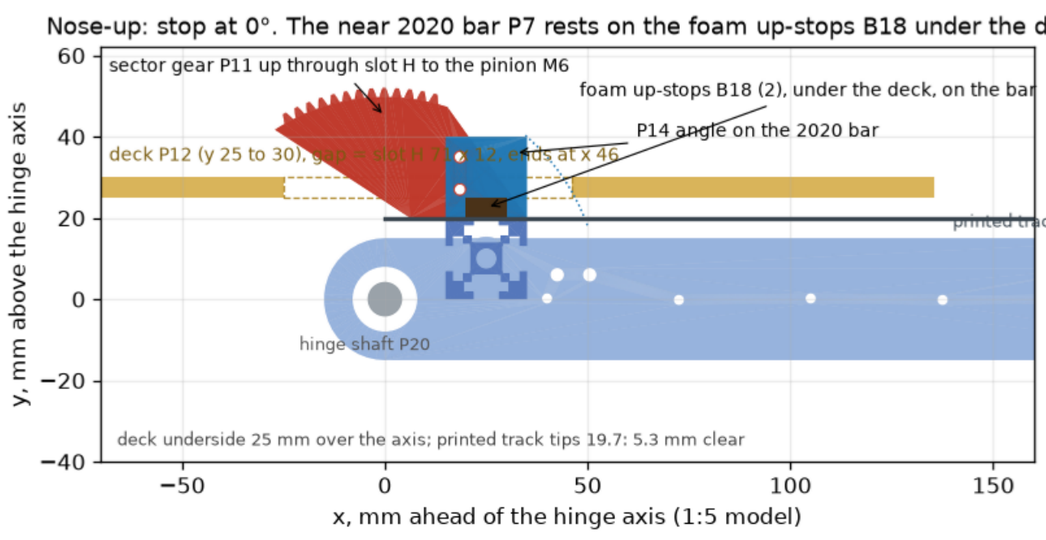
Nhien's Rev B frame (aluminium plates P1 / P2, 2020 and 3030 bars, deck and tray) carrying my printed tracks on bought HTD 5M wheels. Same model as the interactive page.

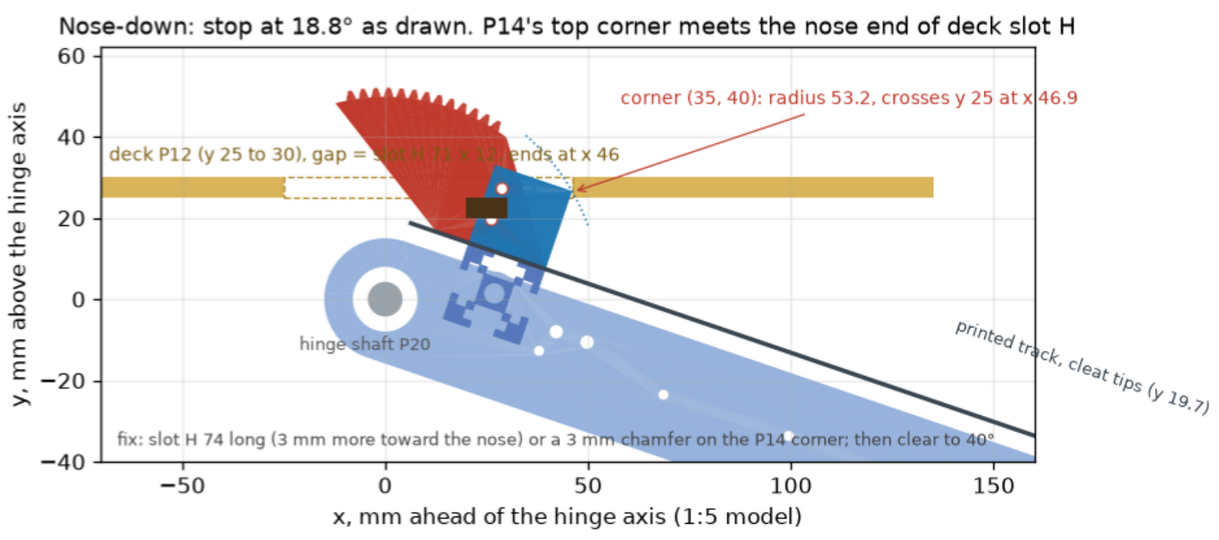


Assembled. Orange 1 to 15 are the joints of the connection map (page 6), each one connection with its own fastener. 16 is the drive pulley on the hinge shaft (the middle sprocket, between the plates), 17 the two drive motors on the deck that belt down to it.



Going down the ES1 stairs: the front segment bent 35° nose-down about the hinge shaft, held by the worm-gear hinge motor; the rear segment flat on the landing. The whole front frame swings: plates P1, both 2020 bars, sector gear, rails, nose idlers and the track.





Where the hinge stops (section at the sector gear, from the shared model, checked part against part at 0.1° steps). Left: it cannot lift. At 0° the near 2020 bar rests on the two foam up-stops B18 that Nhien glues under the deck, and the deck itself runs 135 mm ahead of the hinge with 5 mm over the track. So Rev B bends nose-down only: down the stairs front-first, up the stairs rear-first. Right: nose-down, as drawn, it jams at 18.8°, where the top corner of the sector mount angle P14 (swinging on a 53 mm radius) meets the nose end of the deck's sector slot H, 46 mm ahead of the hinge. With slot H 3 mm longer toward the nose (74 instead of 71), or a 3 mm chamfer on that corner, the swing is clear to 40°. The interactive page stops at both and names them; its 'Deck slot H +3 mm' box shows the fixed version.

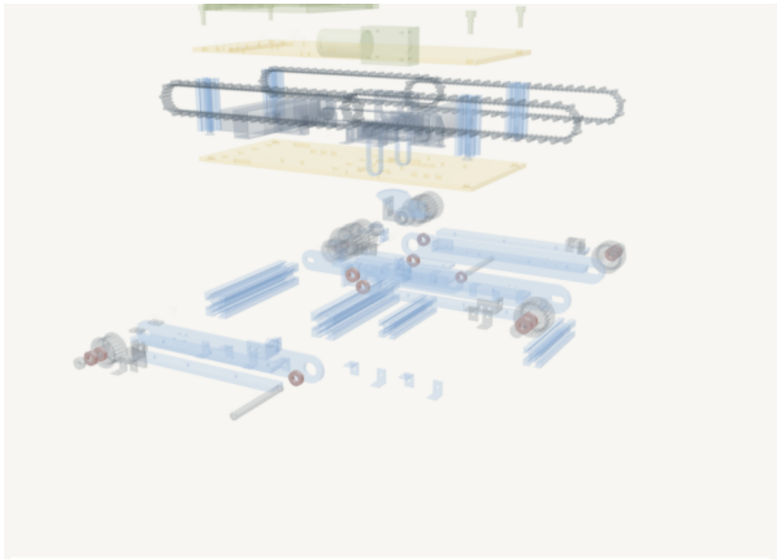
Checks for Nhien

Deck slot H: make it 74 x 12 (3 mm more toward the nose), or chamfer the P14 corner 3 mm. Otherwise the hinge stops at 19° instead of the 35° the stairs need.

Deck clearance over the front track: the deck underside is 25 mm above the axis. My printed track's 3 mm cleats top out at 19.7 mm, 5 mm clear. (The stock belt's 7.9 mm cleats would reach exactly 25 mm and rub.)

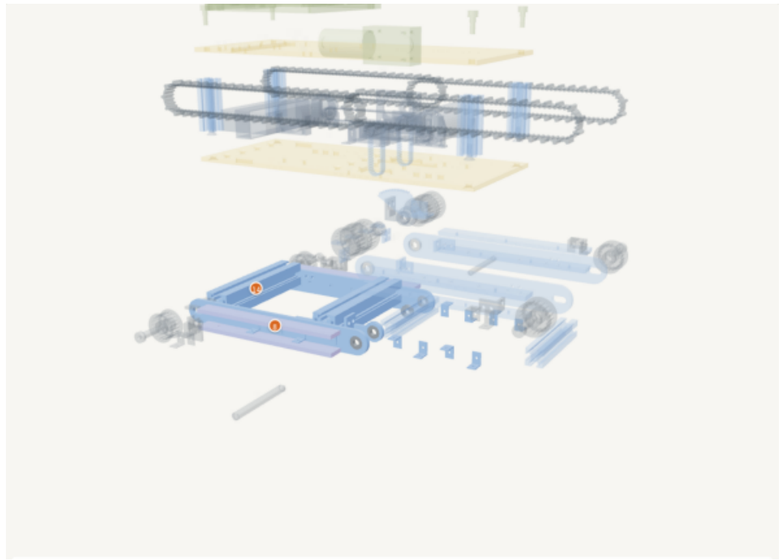
Colours: dark grey = my printed track; light grey = bought wheels, bearings, shafts; blue = Nhien's plates, rails and bars; yellow = deck and tray; slate = motors and belt drive; olive = electronics. Fasteners faded.

2. Piece it together: five of the 11 build steps (slide the interactive page for every step in between)



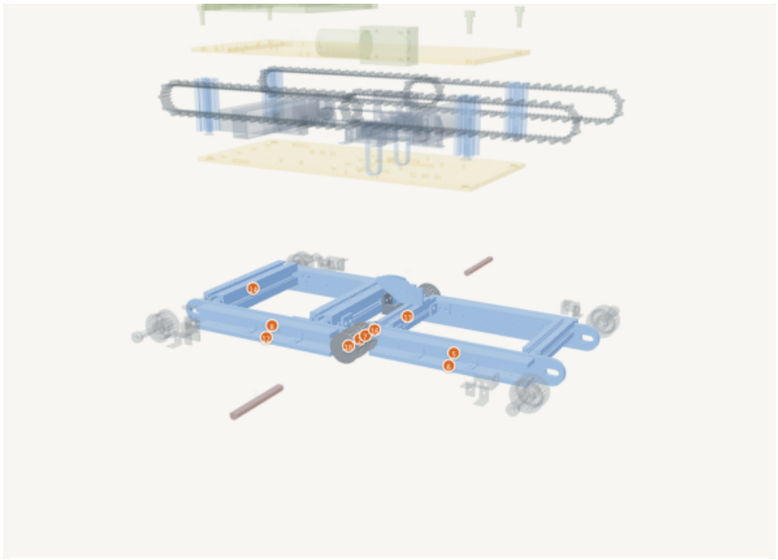
Taken apart

Everything waits in the order it goes on: rear plates and hinge shafts out to the sides, front frame, hinge wheels, deck, tray and tracks stacked above. Step 1: press the 688 bearings into the plates, the P17 supports and the idler wheels.



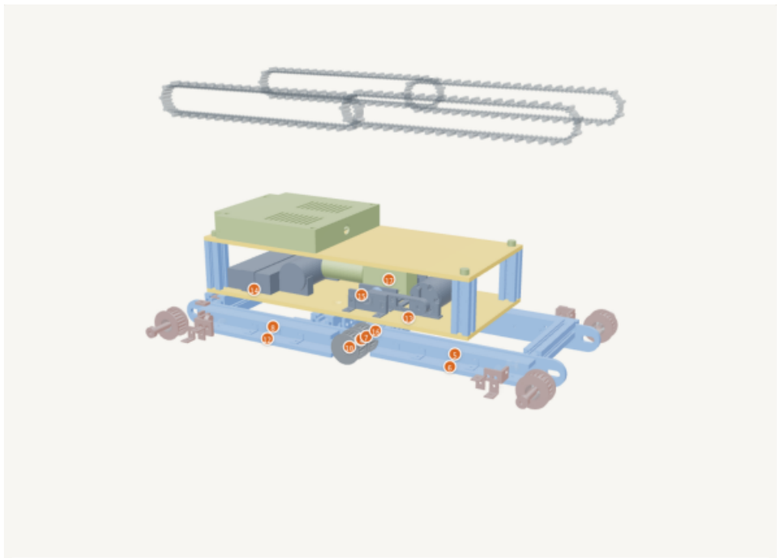
Steps 2 to 3: rear frame

Rear plates P2 onto the 3030 bars with P18 corner brackets (M6x10 + T-nut, 2x M5x8 into the plate) and the P17 hinge supports on the middle bar; then the rear slide rails P5 / P6 onto the outboard faces.



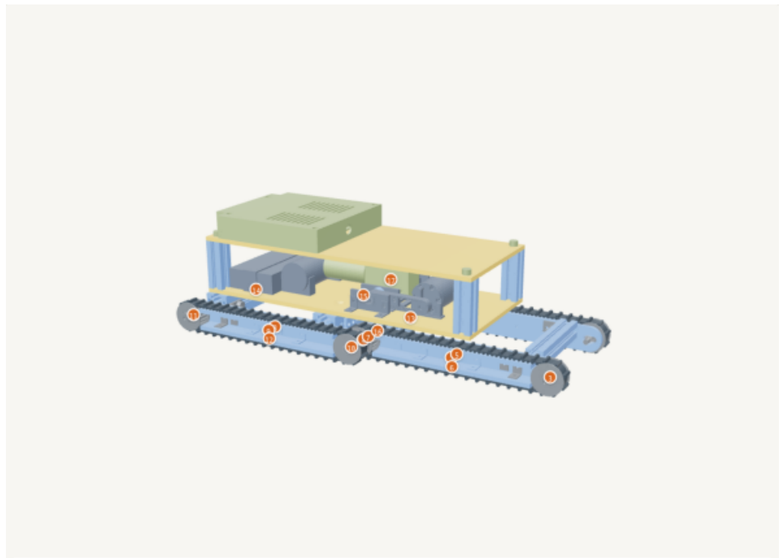
Steps 4 to 7: front frame and hinge

Front plates P1 with the 2020 bars, P19 brackets (M5x8 + T-nut, 2x M4x8) and the sector gear lowered in between the rear plates, rails on. Hinge wheels and drive pulleys dropped in; the hinge shafts push in from outside through rear wheel, rear plate, front wheel, front plate and P17, E-clip and shim at each end. The D-flat drives the wheels; the front pair also gets an M4 grub screw.



Steps 8 to 10: deck and idlers

Deck onto the 3030 bars (8x M5x12 csk + T-nuts), posts, drive motors, belts and hinge motor; tray, controller, battery and cover. Then the idler wheels onto their shoulder screws through the plate slots.



Step 11: tracks on

Each printed track over its two wheels from above; slide the idlers out in the slots to tension and tighten the nyloc nuts. Complete.

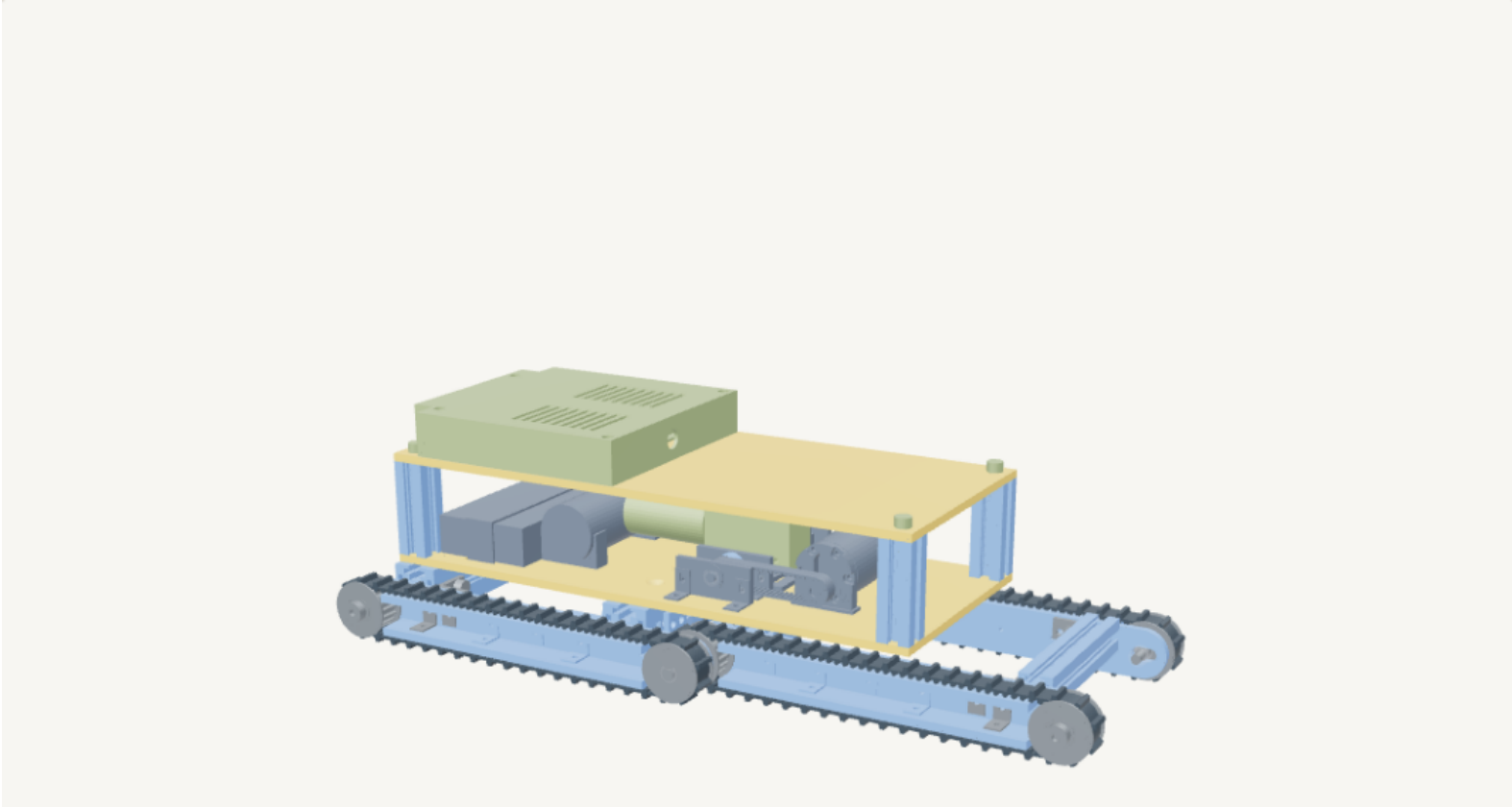
Interactive version

Open the page, drag the Assemble slider left to take the robot apart, then right to rebuild it one connection at a time. Attached parts ride with the part they are fixed to, so nothing passes through anything else. The panel names the joint being made and its fasteners; click any part or number for its details. The Track menu switches between the printed track and the bought belt; Show: drive train only, or frame and tracks only. The Hinge slider stops where the real parts would touch; the stairs toggle puts it on the ES1 flight.

https://claude.ai/artifact/6gnfWidMkD9kDk2BQc1tBw

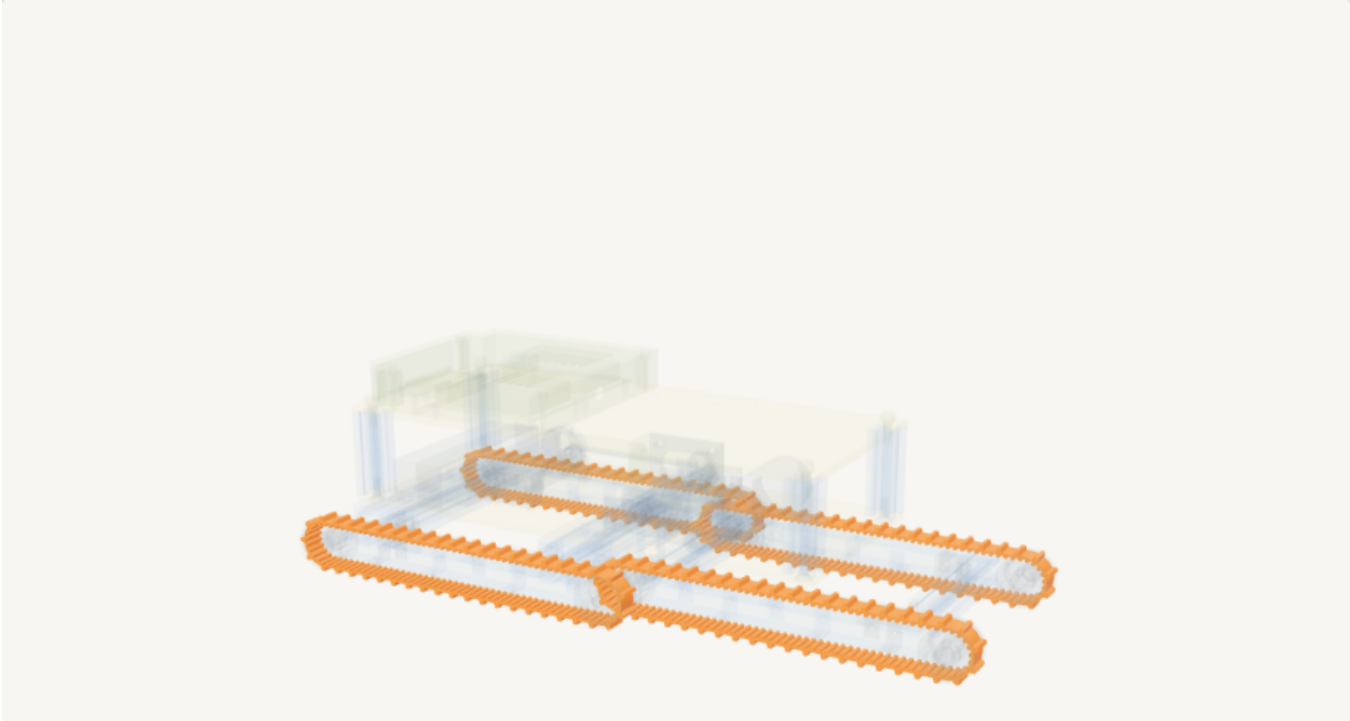
Parts catalogue: my printed track on Nhien's Rev B frame

One page per part. Orange = that part in the shared model of Nhien's Rev B assembly. Blue lines are links: click the vendor line to open the product page. Rev B sheet numbers refer to 'RB4107 Fabrication drawings (Rev B).pdf'.



Mine to make: track x4.
Mine to buy: pulleys x8, TPU.
Nhien's: plates, rails, shafts, bearings, shoulder screws, bars, deck, motors, electronics (his BOM).

Track, printed



Quantity: 4

MAKE: 3D print (mine)

What it does

The printed TPU belt: 110 teeth inside on the toothed wheels, 45 cleats 3 mm high outside, spaced so five span one ES1 step. Fits Rev B's pulleys and 225 mm spacing; its cleat tips sit 5 mm under the deck. The other version of this pack uses Nhien's stock belt with bonded cleats instead.

Where to get it

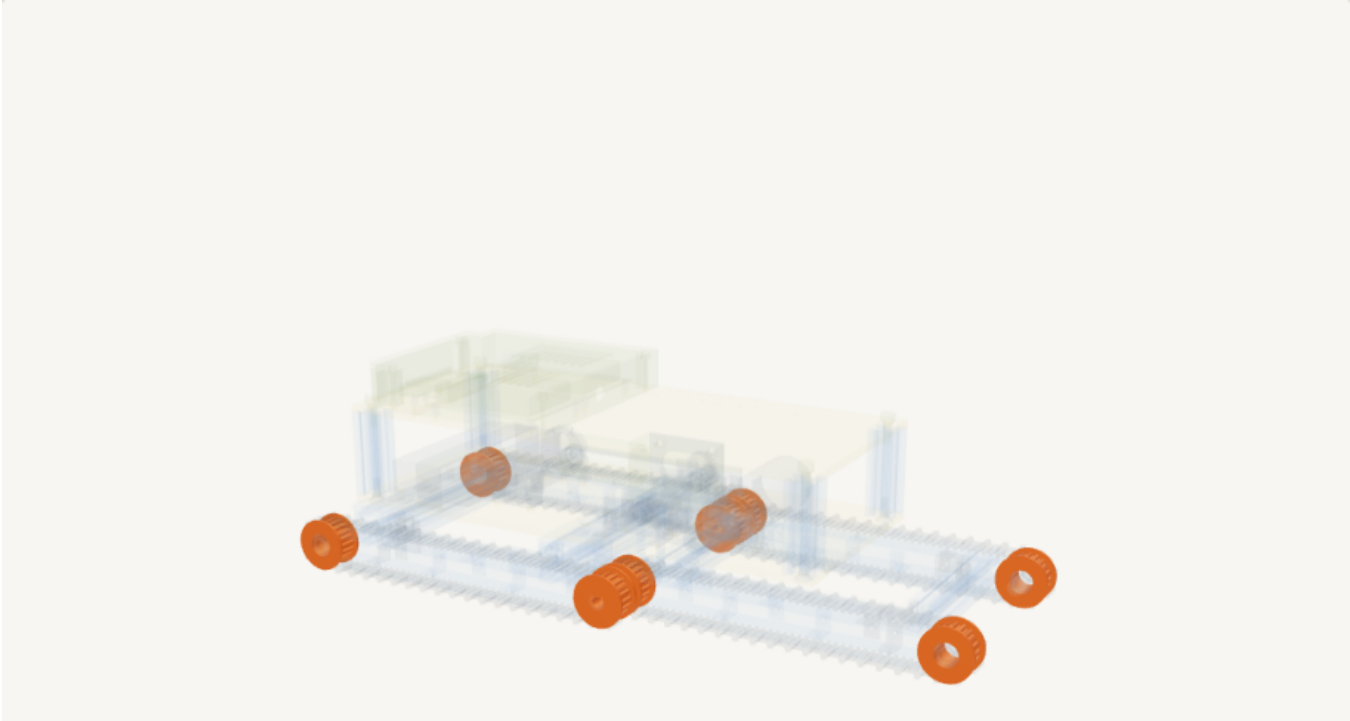
Fab Lab E4-02-06, Tiertime UP300 with the TPU head. File stl/track_TPU_x4.stl, standing as a ring, no supports, 0.2 mm, 100% infill. Filament: Fab Lab stock or Element14 SG TPU 95A (link). Drawing TS-01.

Link: Element14 SG: TPU 95A filament

https://sg.element14.com/multicomp-pro/mc011460/3d-printer-filament-tpu-1-75

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Toothed wheel (timing pulley) HTD 5M 20T



Quantity: 8

BUY plain bore + MODIFY

What it does

Rev B's lanes are only 19-20 mm wide, so there is no room for a set-screw hub. The hinge ones are driven by the D-shaped shaft; the end ones spin on bearings on a shoulder screw.

Where to get it

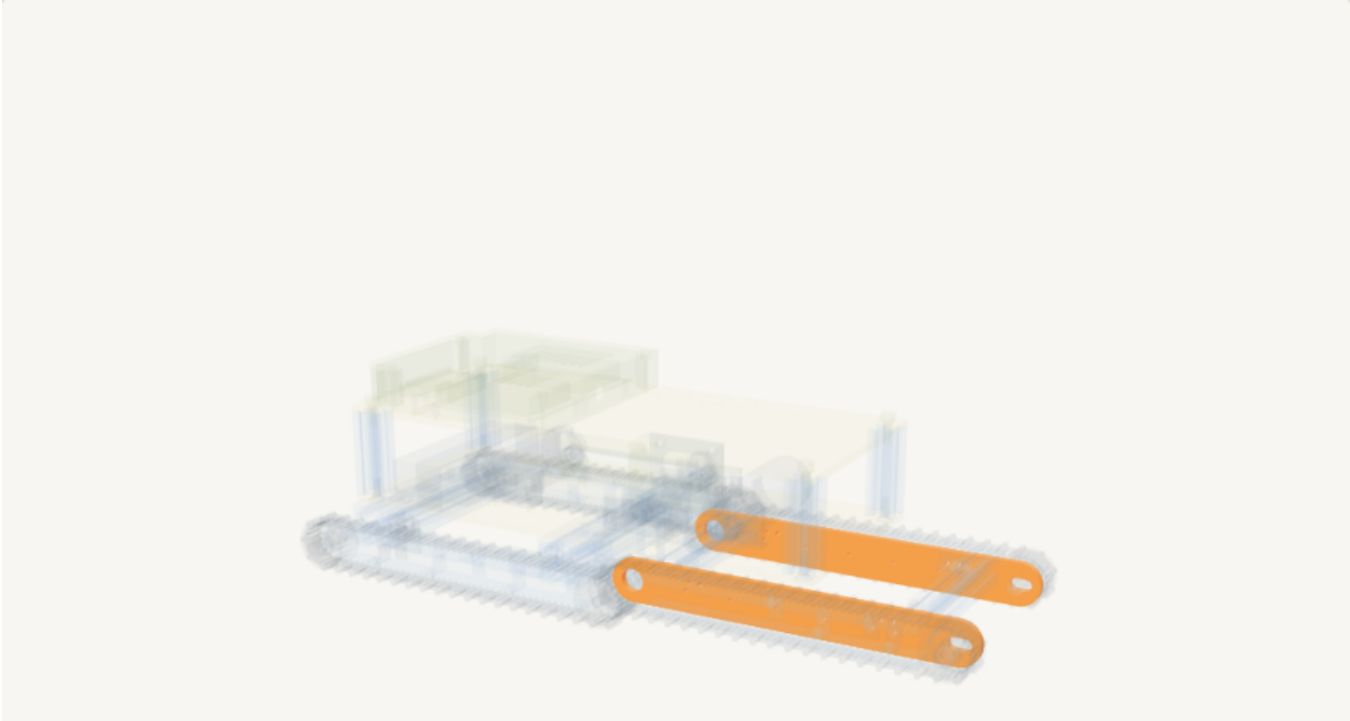
MISUMI SG, Timing Pulleys HTD 5M (link): 20 T, 15 mm belt, flanges both sides, PILOT (plain) bore. Then Professional Workshop: 4 bored to a D-shape Ø8 (hinge shaft, M1, plus an M4 grub on the front pair), 4 bored to Ø16 H7 for two 688 bearings each (end idlers, M2). Rev B sheet 21.

Link: MISUMI SG: HTD 5M timing pulleys

https://sg.misumi-ec.com/vona2/detail/110310532889/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Side plate, front segment (Rev B P1)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

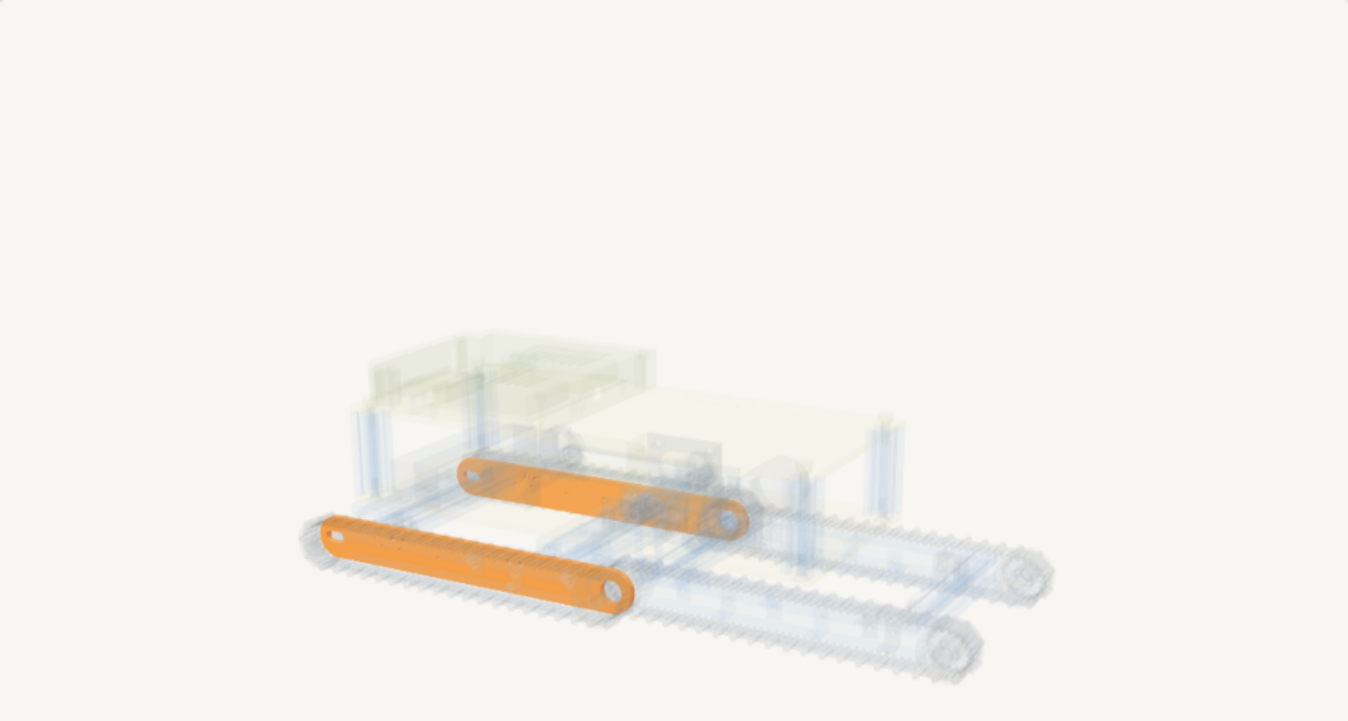
255 x 30.4 plate, inboard (z 55-60). Ø16 hole for the hinge bearing, a slot at the front end for the idler's shoulder screw (slide it to tension the track), tapped holes for the rail brackets and the 2020 bar brackets.

Where to get it

5 mm Al 6061, laser cut from DXF/P1 Side plate - FRONT segment.dxf, then drill and tap per Rev B sheet 4. Fab Lab laser does not cut aluminium: Professional Workshop or an outside laser shop on the vendor list.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Side plate, rear segment (Rev B P2)



Quantity: 2 (1 + 1 opposite hand)

MAKE: laser-cut aluminium (Nhien's)

What it does

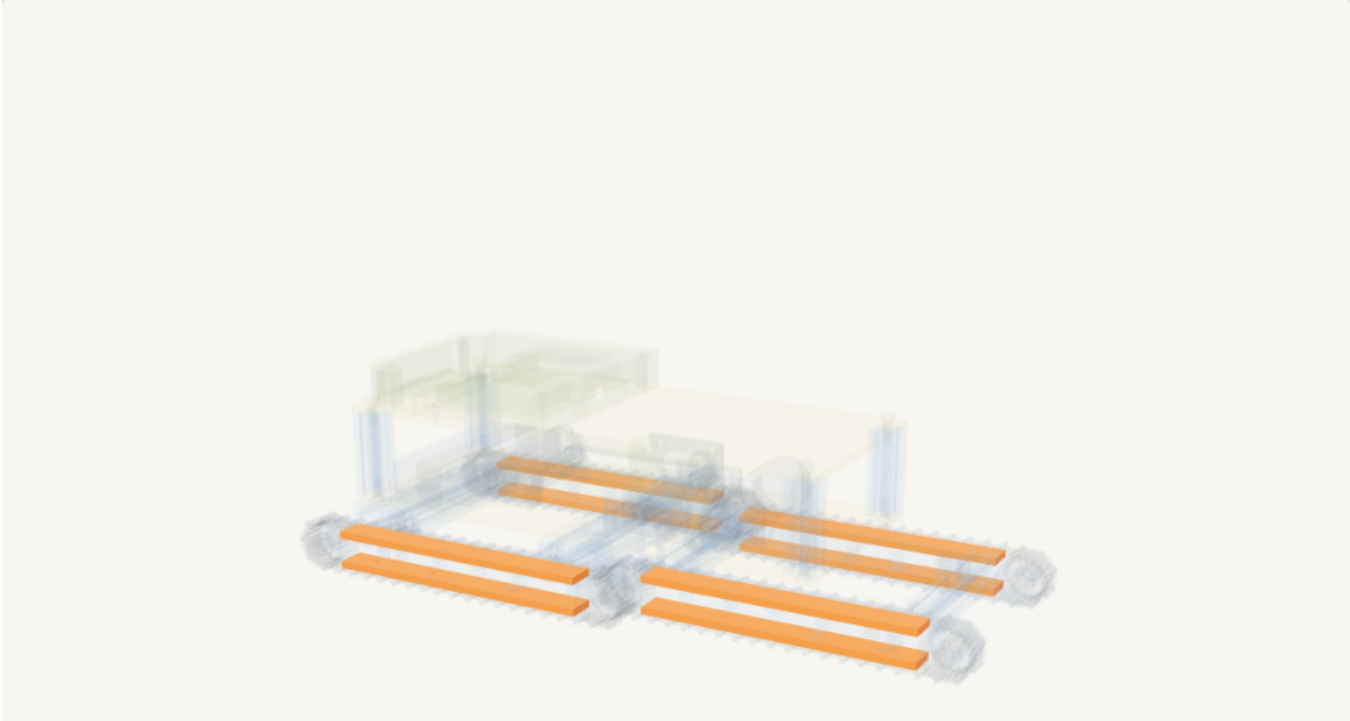
Outboard plate (z 80-85). The rear track runs outside it on a shoulder screw. Holes for the 3030 bar brackets and the rails.

Where to get it

As P1, from DXF/P2 Side plate - REAR segment.dxf, Rev B sheet 5.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Slide rails (Rev B P3-P6)



Quantity: 8

BUY strip + drill

What it does

The strips the track's upper and lower runs slide on so the track cannot sag between the wheels. The track's tooth tips ride on the lower one.

Where to get it

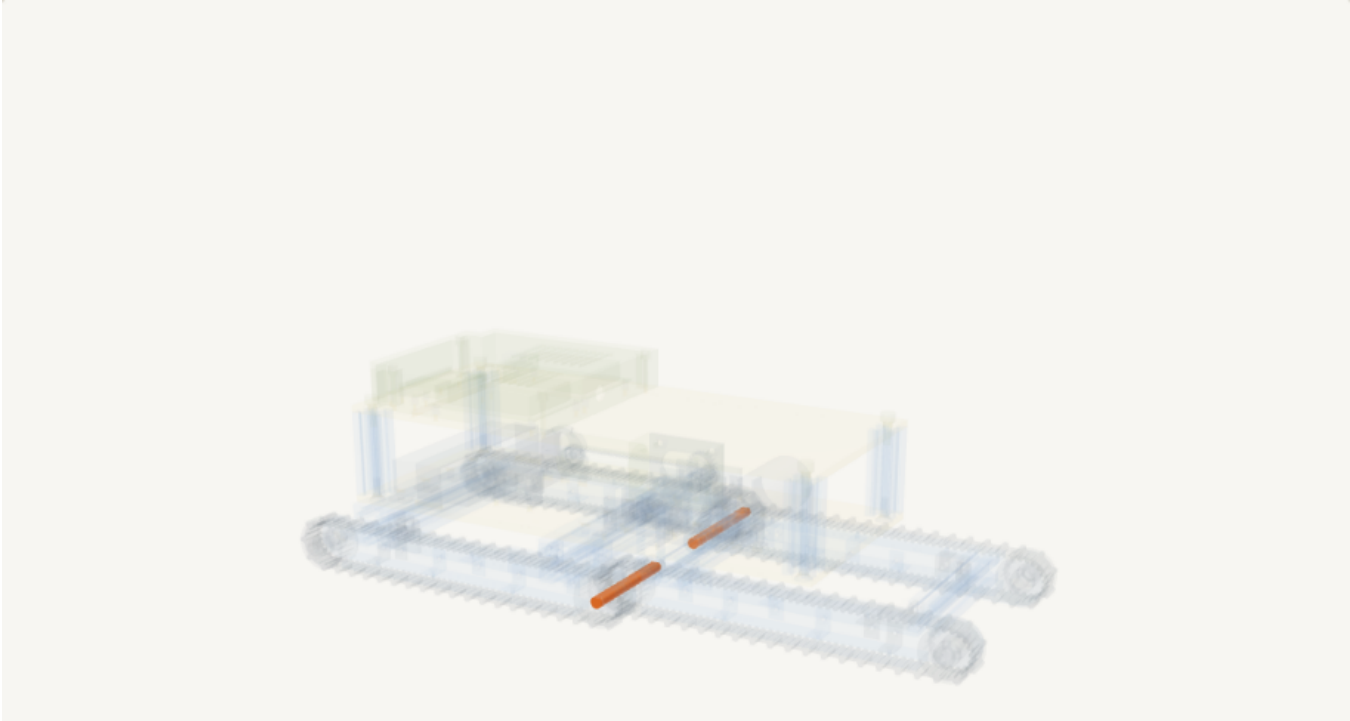
MISUMI SG UHMW-PE strip UPA-189-20-5 (link), 189 x 20 x 5, 3 pilot holes each, Rev B sheet 18. Held to the plates by RS PRO 427-991 angle brackets (RS SG) with M3 screws.

Link: MISUMI SG: plastic strips (UHMW)

https://sg.misumi-ec.com/vona2/mech/M1500000000/M1501000000/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Hinge shaft (Rev B P20 A + B)



Quantity: 2 (or 1 if one motor)

BUY D-shaft, cut

What it does

The hinge axis and the drive: the drive pulleys sit on the D-flat, so no set screws (the front pair gets an M4 grub). Runs in 688 bearings in both plates and in the P17 support.

Where to get it

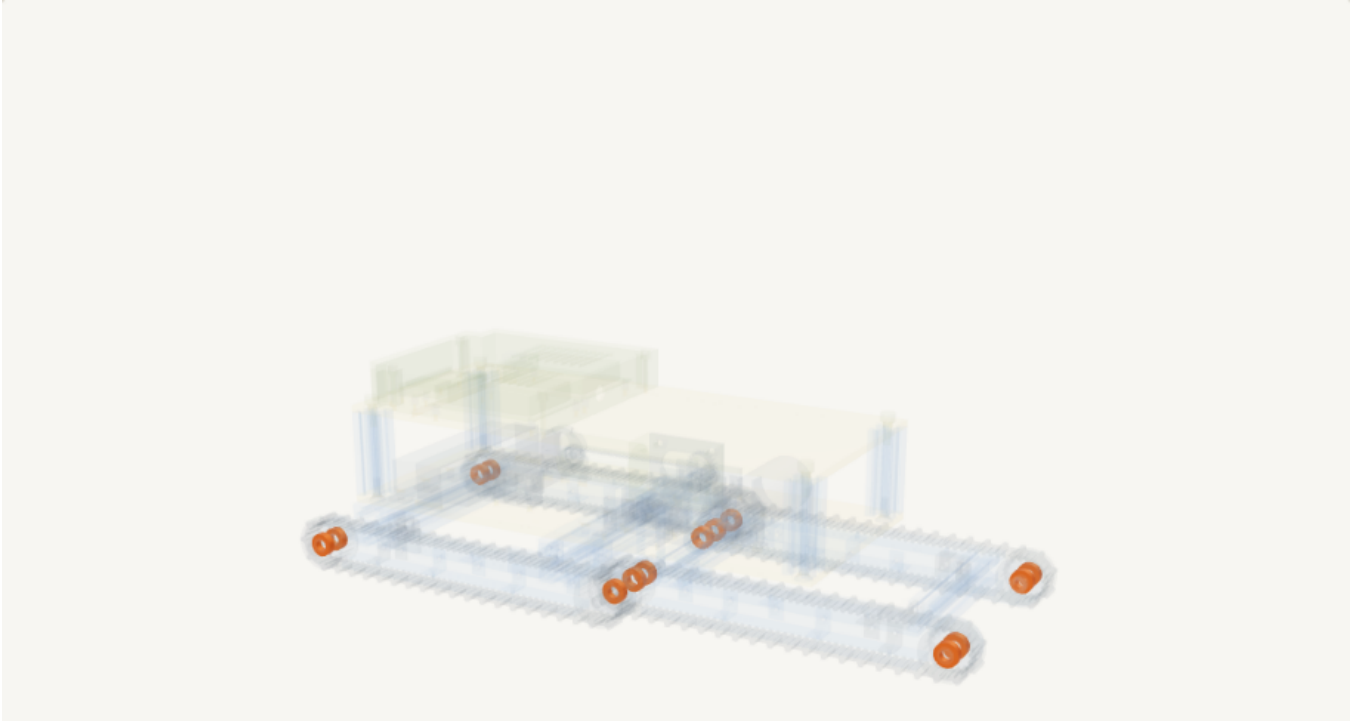
Ø8 h7 D-shaft, flat 7, 200 mm stock (MISUMI SG or RS SG linear shafts, link). Cut to 82.3 (A) and 77.3 (B); E-clip grooves per Rev B sheet 20. One motor per side -> two half shafts; one motor -> one 212 mm shaft.

Link: RS SG: linear shafts

https://sg.rs-online.com/web/c/bearings-seals/linear-motion/linear-shafts/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Ball bearing 688-2RS 8 x 16 x 5



Quantity: 14

BUY: RS Singapore

What it does

Small OD (16) because the plate is only 30 mm tall. Pressed in first, before anything else goes together (build step 1).

Where to get it

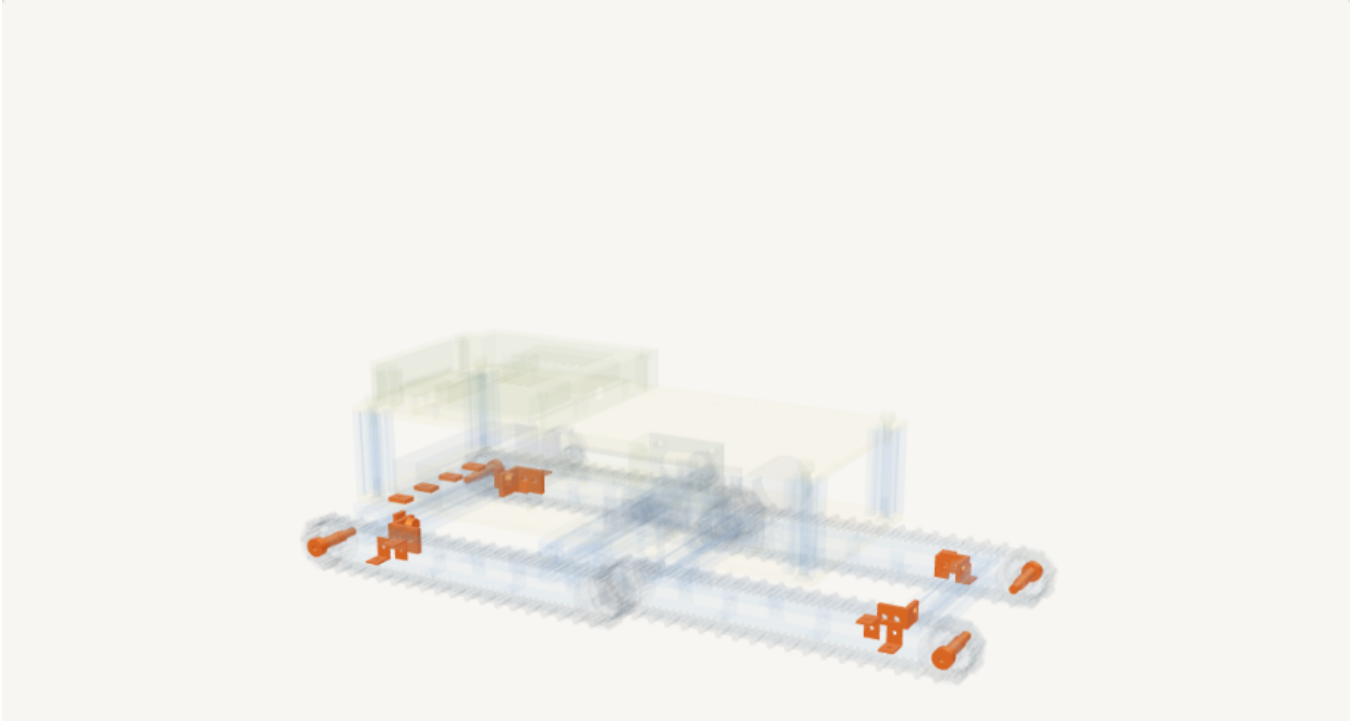
RS SG ball bearings (link): search 688-2RS. 4 in the plate hinge holes, 2 in the P17 supports, 8 inside the four idler pulleys.

Link: RS SG: ball bearings

https://sg.rs-online.com/web/c/bearings-seals/bearings/ball-bearings/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Idler stub axle: shoulder screw Ø8 x 25, M6



Quantity: 4

BUY: MISUMI SG or RS SG

What it does

One per plate end, through the slot; the end wheel spins on two 688s on its shoulder. Slide outwards in the slot and tighten the nut to tension the track.

Where to get it

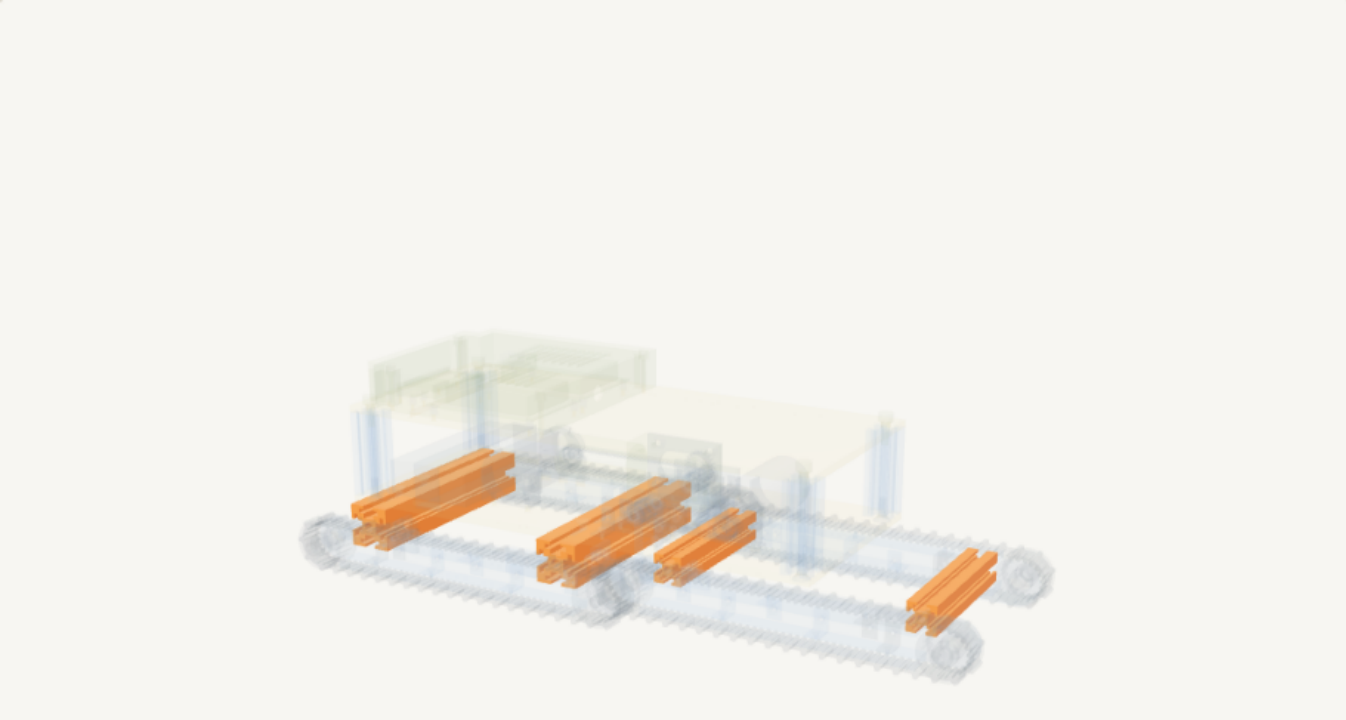
ISO 7379 8 x 25 shoulder screw (link), with 8 x 11 x 1 washer, 8 x 11 x 0.5 shim, 8 x 10 x 9 spacer, M6 washer and nyloc. Rev B sheet 3, idlers.

Link: RS SG: shoulder screws

https://sg.rs-online.com/web/c/fasteners-fixings/screws-bolts/shoulder-screws/

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Cross bars: 2020 x 110 and 3030 x 160 (Rev B P7, P8)



Quantity: 2 + 2

BUY profile, cut

What it does

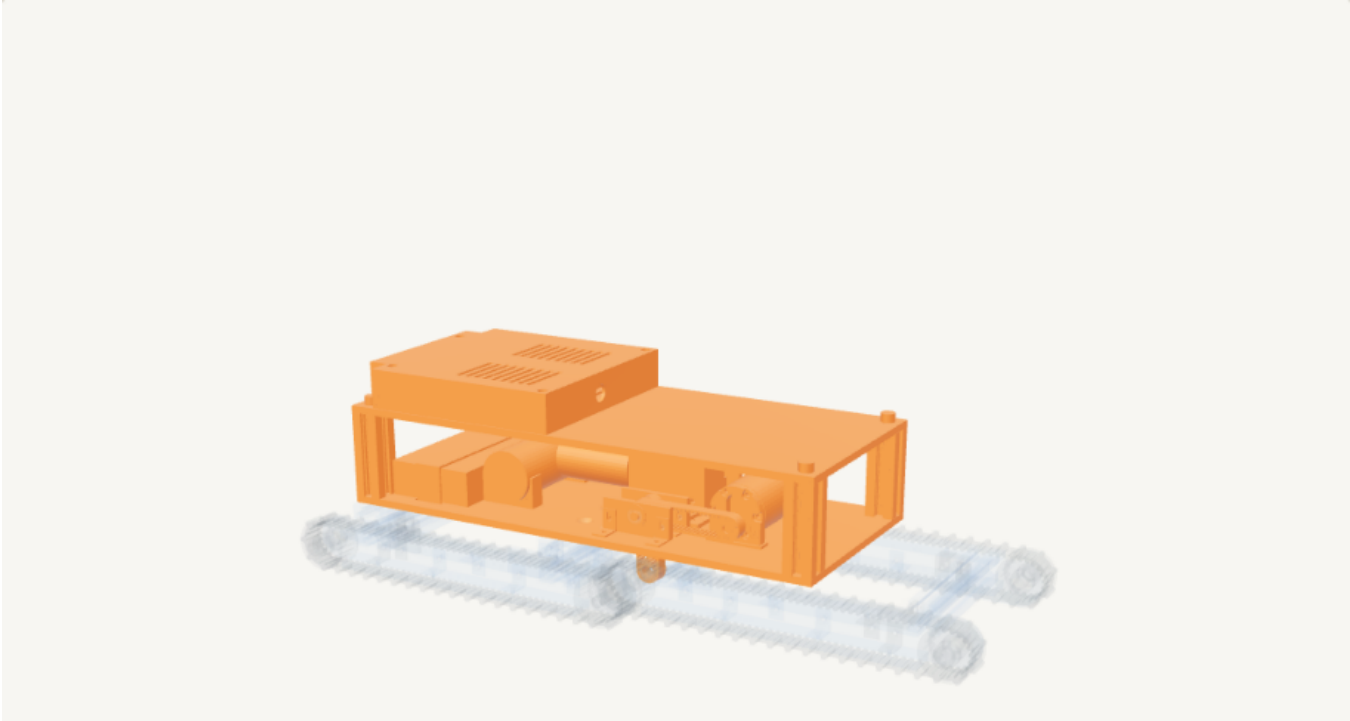
They hold the plates apart and carry the deck. The 2020 bars belong to the front segment and swing with it; the 3030 bars are what the deck bolts to. The near 2020 bar also carries the sector gear angle P14 and rests on the foam up-stops at 0°.

Where to get it

2020 B-type slot 6 (1 m) and 3030 slot 8 (0.5 m) from MISUMI SG or a local supplier; cut per Rev B sheet 19. Corner brackets P18/P19 from aluminium angle, T-nuts M5/M6.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Deck and tray (Rev B P12, P16) and everything on them



Quantity: 1 + 1

Nhien's parts

What it does

The 'box': 160 mm wide between the rear plates. Motors, drivers, battery and the platform mount sit here. Its edges overhang the front track lanes by 12 mm at 25 mm above the axis; slot H lets the sector gear through.

Where to get it

Al 5 mm laser cut, Rev B sheets 6-8; posts P15, controller cover P22, motors, electronics per the Rev B BOM.

Orange = this part in the Rev B assembly; everything else faded. 1:5 model, same model as the interactive page.

Procurement, Rev B build with my printed track

Date 15 Oct 2026. Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien's pack (`RB4107 Rev B`), whose BOM covers everything in section 2.

1. Mine (track side)

Part	Qty	Make / buy	Where	Price	Arrives
Track, TPU 95A, `stl/track_TPU_x4.stl`	4	**Make**: Fab Lab UP300 (TPU head)	Element14 SG TPU 95A filament if the lab has none	~S$45/kg, uses 130 g	filament 1-2 days; 3 h print each
Fit-test slice, `stl/fit_test_track_5mm_TPU.stl`	1	Make	same	-	10 min
Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl`	1	Make, PLA	any printer	-	1 h
Timing pulley HTD 5M 20T, 15 mm belt, flanged, **plain (pilot) bore**	8	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x D-bore Ø8 (M1), 4 x Ø16 H7 (M2)	8	**Make**: Professional Workshop request, Rev B sheet 21	E3-01-06	free	workshop queue, allow 1-2 weeks

2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
Rail brackets RS PRO 427-991 (angle 15 x 14.6 x 10)	32	Buy	RS SG stock no. 427-991
Hinge shafts P20 A 82.3 + B 77.3 (or one 212), Ø8 D-shaft flat 7	2	Buy stock, cut	RS SG linear shafts or MISUMI SG
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets P18 (25x25x4 angle) x4, P19 (20x20x3) x4, T-nuts, E-clips, shims	-	Buy / make	Rev B BOM sections 3-4
Deck P12, tray P16, cover P22, motors, drivers, ESP32, battery	-	Theirs	Rev B BOM

3. What is no longer needed from my earlier list

Printed side plates (4), printed spacers (6), M4 rods and nuts, Ø8 plain rod for three axles, 608ZZ x10, M4 set screws. Keep the STL files in case the team falls back to the all-printed frame.

4. Order of events

1. Order the 8 plain-bore pulleys today (critical path, then the workshop bore mod); print the stand-in pulley and fit-test slice; print the 4 tracks as soon as a UP300 slot is free.
2. Nhien: laser plates, order rails, bearings, shoulder screws, profiles.
3. Workshop: pulley bores, plate tapping, shaft cutting (one request form).
4. Assemble per Rev B sheet 3; fit my tracks on the real pulleys; tension at the slots.

Layout with Nhien's side plates (Rev B P1 / P2): what changes on the track side

Read from `RB4107 Rev B/STEP` (plates P1, P2 and the full assembly, 15 Oct 2026). Rev B's frame origin is the hinge axis with the A side at z < 60; below everything is converted to **z from the vehicle centre line**, one side (the other mirrors), mm at 1:5.

Their plates

	Rev B plate	My printed plate (for comparison)
Material	5 mm aluminium 6061, laser cut, holes tapped	7 mm PETG, printed
Size	255 x 30.4 (R15 ends)	265 x 45 (R20 ends)
Axle spacing	225 (hinge hole to the middle of the idler slot)	225
Hinge hole	Ø16 for a 688-2RS bearing (8 x 16 x 5)	Ø22.2 for a 608ZZ
End hole	slot 220.4-232.6 for an Ø8 shoulder-screw stub axle: slide to tension the track	round pocket
Rail	separate UHMW strips P3-P6, screwed on through the C holes with small brackets	printed as part of the plate
Cross bars	2020 profile x 110 (front), 3030 profile x 160 (rear), corner brackets + T-nuts	M4 rod + printed spacer

Where everything sits across the width (per side, from the centre line)

z	What
0 to 24 (A side) / 29 (B side)	free: hinge shaft ends here, hinge pinion / sector gear region
36 to 41	P17 hinge-shaft bearing support (688 bearing), bolted to the middle 3030 bar
55 to 60	**front plate P1** (688 bearing in it at the hinge)
60.5 to 79.5	front segment track lane: pulley 19 over flanges, track 62.5 to 77.5
80 to 85	**rear plate P2** (688 bearing at the hinge)
85.5 to 104.5	rear segment track lane, **outboard of the rear plate**; track 87.5 to 102.5
106	end of the hinge shaft

So: 205 over the track faces, 209 over the pulley flanges, and the frame between the rear plates is 160 wide inside (not 80 as in my layout): the box can be up to 160 wide.

Consequences for the parts I was making

Part	With their plates

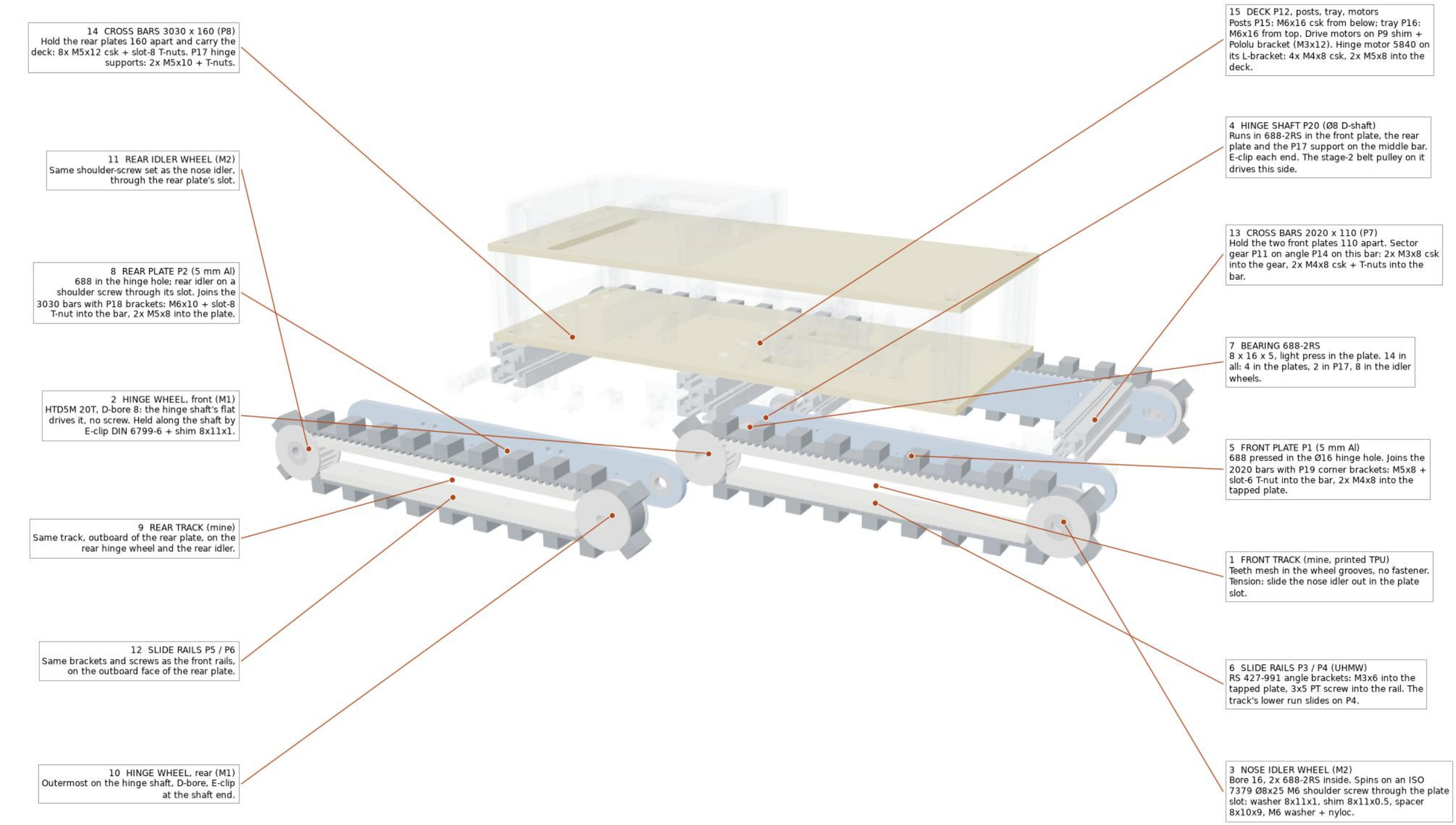
Track x4	unchanged: 550 loop, 110 teeth, 15 wide, 45 cleats. Still printed. The only printed part left.
Side plates	theirs (P1, P2). Not printed.
Spacers	**none.** The 2020 and 3030 bars hold the plates apart.
Cross bar rods, nuts, washers	**none.** Replaced by the bars, P18/P19 corner brackets, T-nuts.
Middle axle 230	replaced by Rev B's two hinge half-shafts P20: Ø8 D-shaft (flat 7), A = 82.3 long (z 14 to 106 on the A side), B = 77.3 (z 31 to 106 on the B side). Two halves because Rev B has one motor per side. If the team goes to **one motor**, use one Ø8 D-shaft 212 long (z -106 to +106) instead, same stock.
Front and rear axles	**none.** The end pulleys ride on Ø8 x 25 M6 shoulder screws (ISO 7379) through the plate slots, one per plate, 4 in all, each with two 688-2RS inside a Ø16-bored pulley.
Bearings	608ZZ x10 -> **688-2RS x14** (8 x 16 x 5): 4 in the plate hinge holes, 2 in the P17 supports, 8 in the idler pulleys.
Pulleys	same HTD 5M 20T 15 mm belt, but the lanes are only 19-20 wide, so there is **no room for a set-screw hub**. Rev B modifies them: the 4 on the hinge shaft get a D-bore 8 (M1), the 4 idlers get a Ø16 H7 bore for the 688s (M2). That is lathe work at the Professional Workshop; order plain-bore (pilot-bore) pulleys for it.
Set screws	none on the pulleys (D-bore drives them); E-clips DIN 6799-6 and a shim on each shaft end instead.

What I need from Nhien to finish the hybrid

1. One motor or two? It decides whether the hinge shaft is one piece (212) or two (82.3 + 77.3).
2. Confirm the rail strips P3-P6 are coming (MISUMI UPA-189-20-5 UHMW, 8 off) or whether the lower run just rides on nothing; my track's tooth tips sit 13.34 below the axle and need the rail.
3. The track is still mine to print; the step-matched cleat pitch (12.22) replaces Rev B's 22 cleats at 25 mm on a stock belt. Both fit the same pulleys, so either can be tested on their frame.

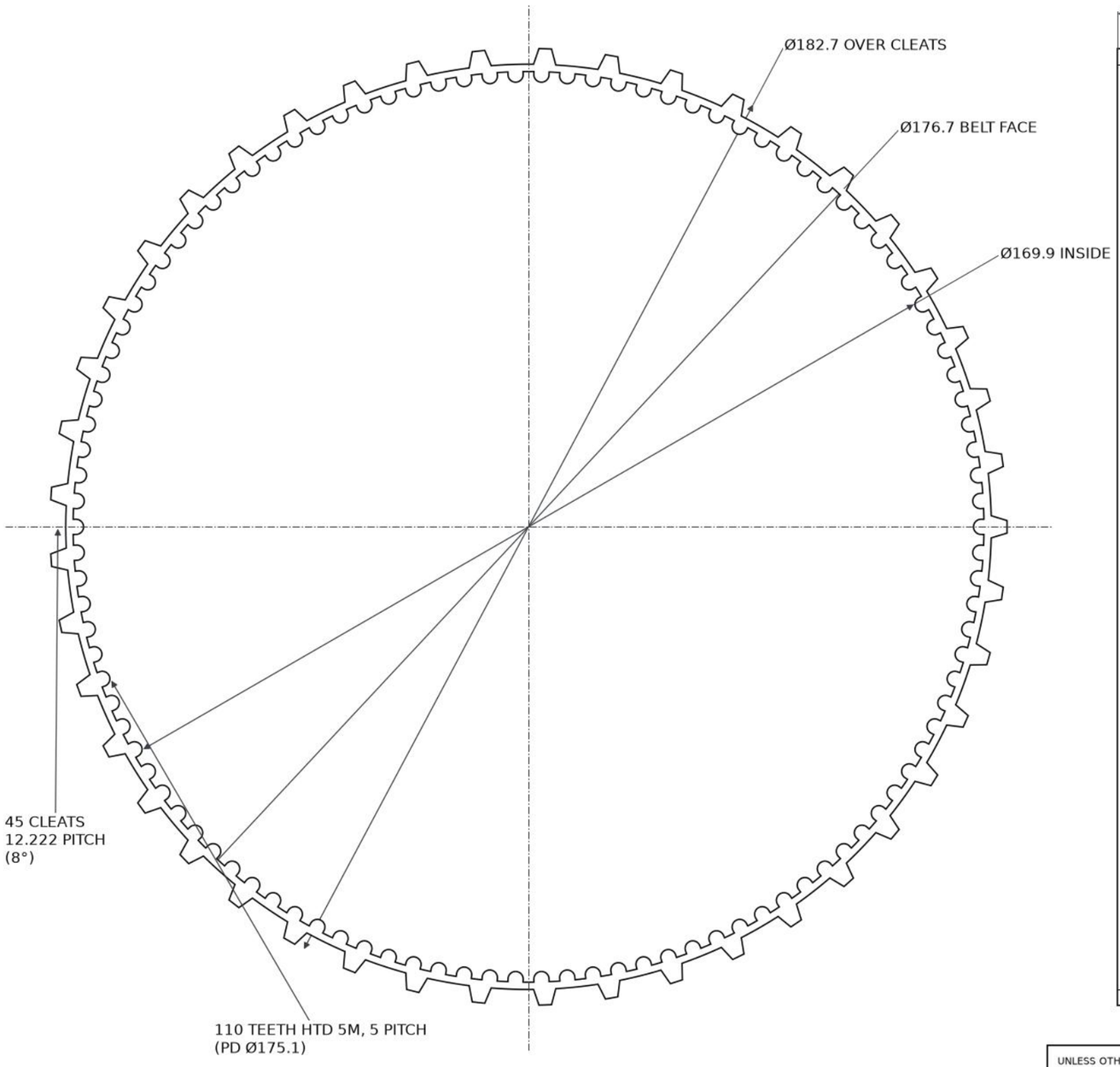
How it all connects: one side pulled apart, every joint and its fastener

Nhien's Rev B frame with my printed track. Near side exploded outwards from the hinge shaft (plate, then track and wheels, then rear plate, then rear track); far side left assembled. Bought parts grey, Nhien's blue, deck yellow, electronics ghosted.



Assembly order (Rev B sheet 3): 1 bearings into plates, P17 and idlers 2 rear frame: P2 + 3030 bars + P18 3 front frame: P1 + 2020 bars + P19 + sector gear 4 hinge shaft through rear wheel, rear plate, front wheel, front plate, P17; E-clips 5 deck, posts, tray 6 motors, belts, tension 7 idlers on shoulder screws, tension the tracks 8 electronics, cover. Fasteners from Rev B sheet 3; my track and its fit test from this pack.

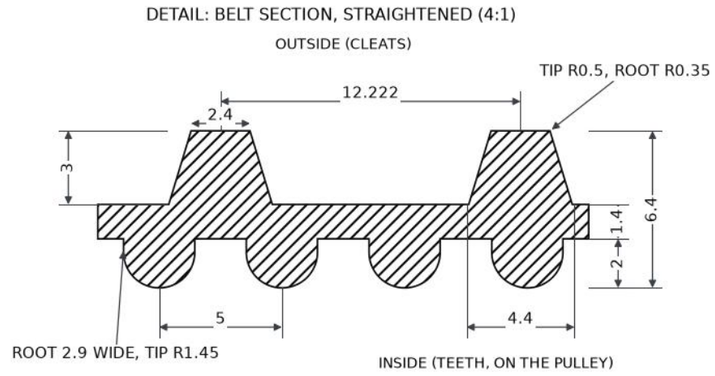
FRONT VIEW



EDGE VIEW



DETAIL: BELT SECTION, STRAIGHTENED (4:1)



NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
TRACK, 110T HTD 5M, 45 CLEATS

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-01

QTY.: 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

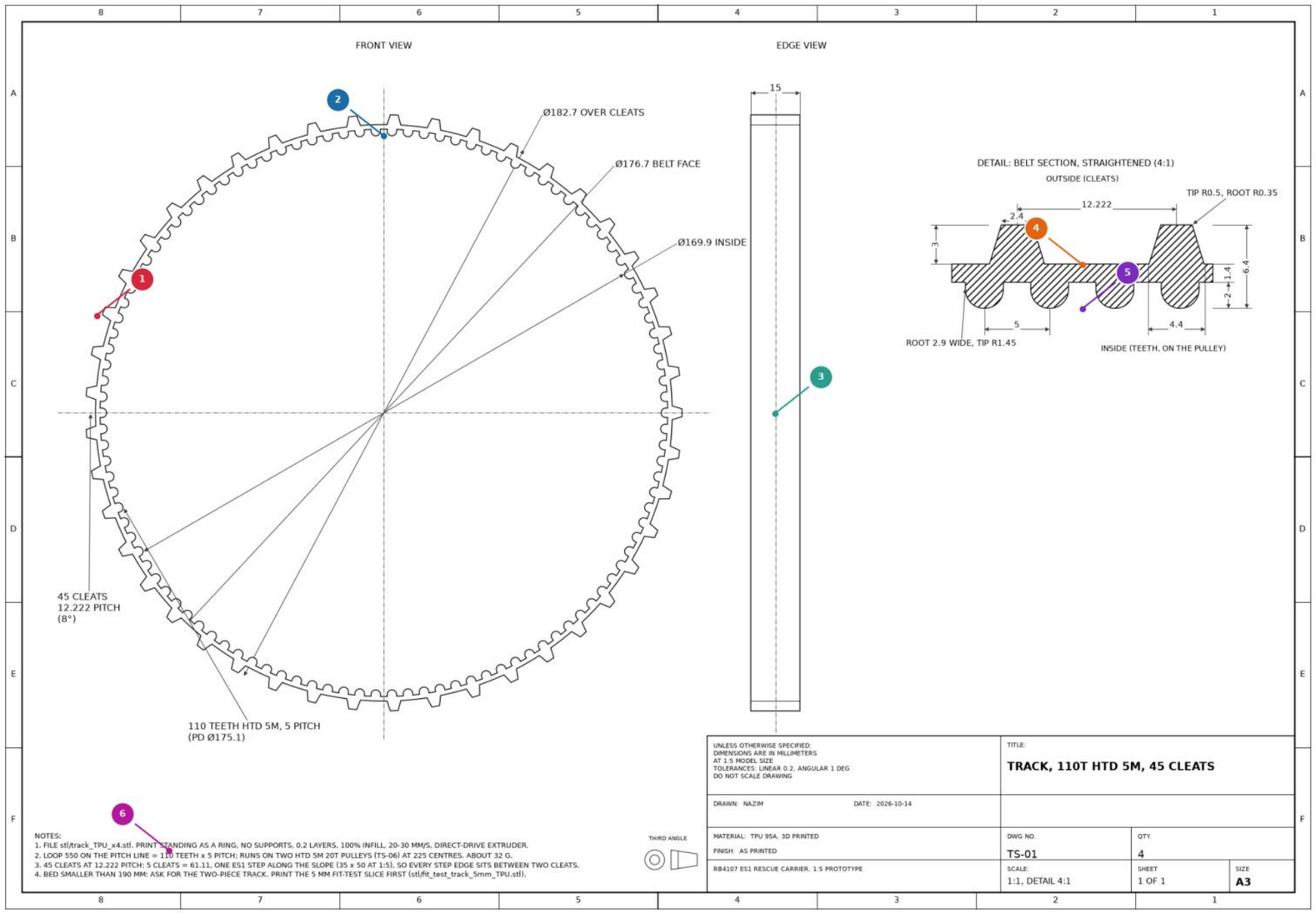
SCALE: 1:1, DETAIL 4:1

SHEET: 1 OF 1

SIZE: **A3**

How to read TS-01: the track

Markers below sit on a reduced copy of sheet TS-01 (previous page, full size). Numbers match the key on the right.



1 Cleats, outside

45 bumps 12.22 apart (8 degrees). Five of them span one ES1 step, so every step edge lands between two cleats and the track holds like a gear on a rack.

2 Teeth, inside

110 HTD 5M teeth, 5 apart. These mesh with the 20-tooth wheels; this is what stops the track slipping.

3 Edge view

The ring seen from the side: 15 wide, 182.7 tall over the cleats.

4 Cleat profile (4:1)

A cleat is 3 tall, 4.4 wide at its base and 2.4 at the tip, with rounded corners so it releases from the step.

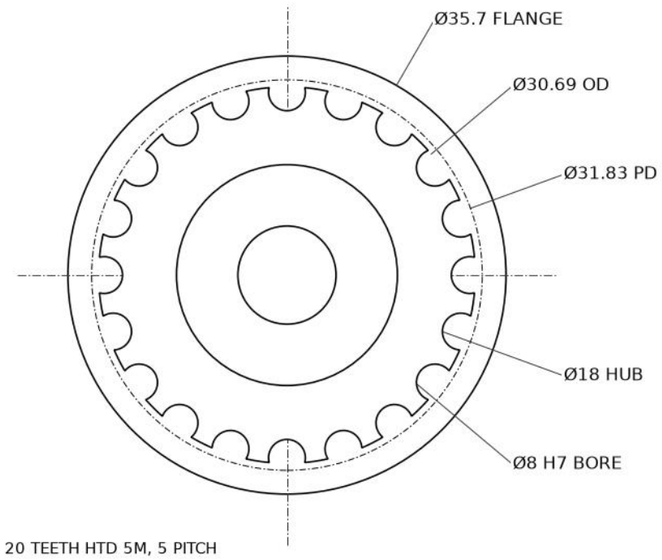
5 Tooth profile (4:1)

A tooth is 2 deep with a 1.45 radius tip and a 2.9 root; the 1.4 backing between teeth and cleats is the belt itself. Whole section 6.4 thick.

6 Notes

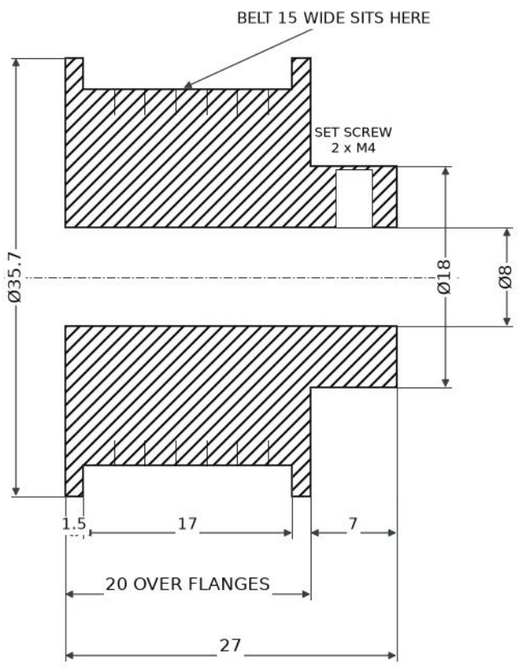
Print standing up as a ring, no supports, 100% infill, TPU 95A, on the Fab Lab UP300. Four tracks, about 32 g each.

FRONT VIEW



20 TEETH HTD 5M, 5 PITCH

SECTION (ALONG THE AXLE)



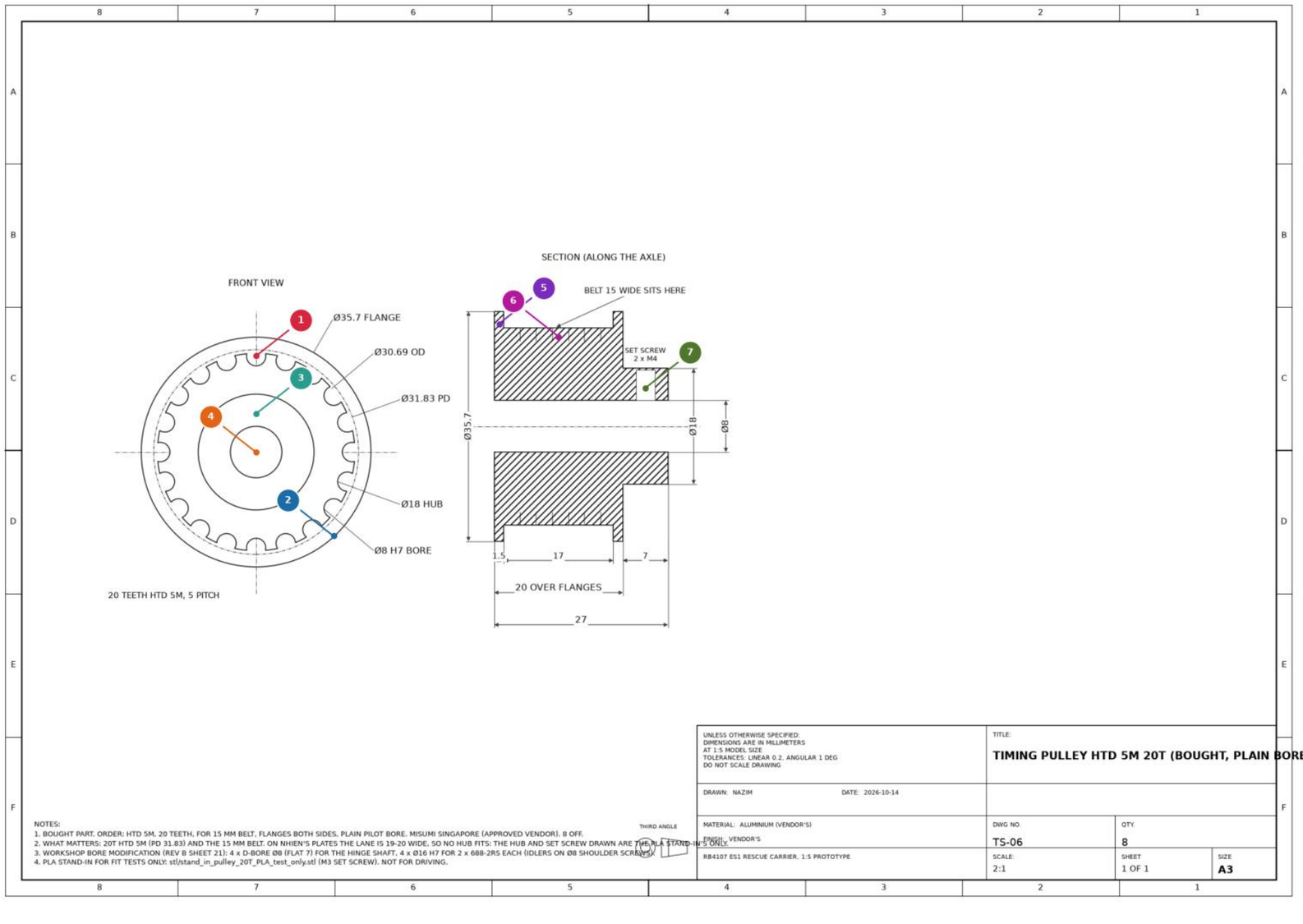
NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, FLANGES BOTH SIDES, PLAIN PILOT BORE. MISUMI SINGAPORE (APPROVED VENDOR). 8 OFF.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83) AND THE 15 MM BELT. ON NHIEN'S PLATES THE LANE IS 19-20 WIDE, SO NO HUB FITS: THE HUB AND SET SCREW DRAWN ARE THE PLA STAND-IN'S ONLY.
3. WORKSHOP BORE MODIFICATION (REV B SHEET 21): 4 x D-BORE Ø8 (FLAT 7) FOR THE HINGE SHAFT, 4 x Ø16 H7 FOR 2 x 688-2RS EACH (IDLERS ON Ø8 SHOULDER SCREWS).
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS AT 1:5 MODEL SIZE TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG DO NOT SCALE DRAWING	TITLE: **TIMING PULLEY HTD 5M 20T (BOUGHT, PLAIN BORE)**		
DRAWN: NAZIM DATE: 2026-10-14			
MATERIAL: ALUMINIUM (VENDOR'S) FINISH: VENDOR'S	DWG NO. TS-06	QTY. 8	
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE	SCALE: 2:1	SHEET 1 OF 1	SIZE **A3**

How to read TS-06: the toothed wheel (bought, plain bore)

Markers below sit on a reduced copy of sheet TS-06 (previous page, full size). Numbers match the key on the right.



1 — 20 grooves

The track's teeth drop into these. HTD 5M profile, 5 pitch, pitch diameter 31.83.

2 — Flange

A thin Ø35.7 disc on each side of the teeth. Keeps the 15 wide track from walking off sideways.

3 — Hub (stand-in only)

On Nhien's plates there is no room for a hub: the real wheels are plain-bore and get a D-bore or a Ø16 bearing bore at the workshop.

4 — Bore

Ordered plain. Workshop: 4 x D-bore Ø8 (driven by the hinge shaft's flat), 4 x Ø16 H7 (two 688 bearings, idlers on shoulder screws).

5 — Flange, in section

1.5 thick, both sides. Total 20 over the flanges, which is the lane width in the layout.

6 — Belt seat

17 between the flanges; the 15 wide track sits here with 1 each side.

7 — Hub and set screws (stand-in only)

Not used on the Rev B frame: the D-bore drives the wheel, E-clips locate it. Only the 20 over the flanges matters for the lane width.

NOTES:
1. BOUGHT PART. ORDER: HTD 5M, 20 TEETH, FOR 15 MM BELT, FLANGES BOTH SIDES, PLAIN PILOT BORE. MISUMI SINGAPORE (APPROVED VENDOR). 8 OFF.
2. WHAT MATTERS: 20T HTD 5M (PD 31.83) AND THE 15 MM BELT. ON NHIEN'S PLATES THE LANE IS 19-20 WIDE, SO NO HUB FITS: THE HUB AND SET SCREW DRAWN ARE THE PLA STAND-IN'S ONLY.
3. WORKSHOP BORE MODIFICATION (REV B SHEET 21): 4 x D-BORE Ø8 (FLAT 7) FOR THE HINGE SHAFT, 4 x Ø16 H7 FOR 2 x 688-2RS EACH (IDLERS ON Ø8 SHOULDER SCREWS).
4. PLA STAND-IN FOR FIT TESTS ONLY: stl/stand_in_pulley_20T_PLA_test_only.stl (M3 SET SCREW). NOT FOR DRIVING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS AT 1:5 MODEL SIZE. TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG. DO NOT SCALE DRAWING.	TITLE: TIMING PULLEY HTD 5M 20T (BOUGHT, PLAIN BORE)		
DRAWN: NAZIM DATE: 2026-10-14			
MATERIAL: ALUMINIUM (VENDOR'S) FINISH: VENDOR'S	DWG NO. TS-06	QTY. 8	
RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE	SCALE: 2:1	SHEET 1 OF 1	SIZE A3

PART B

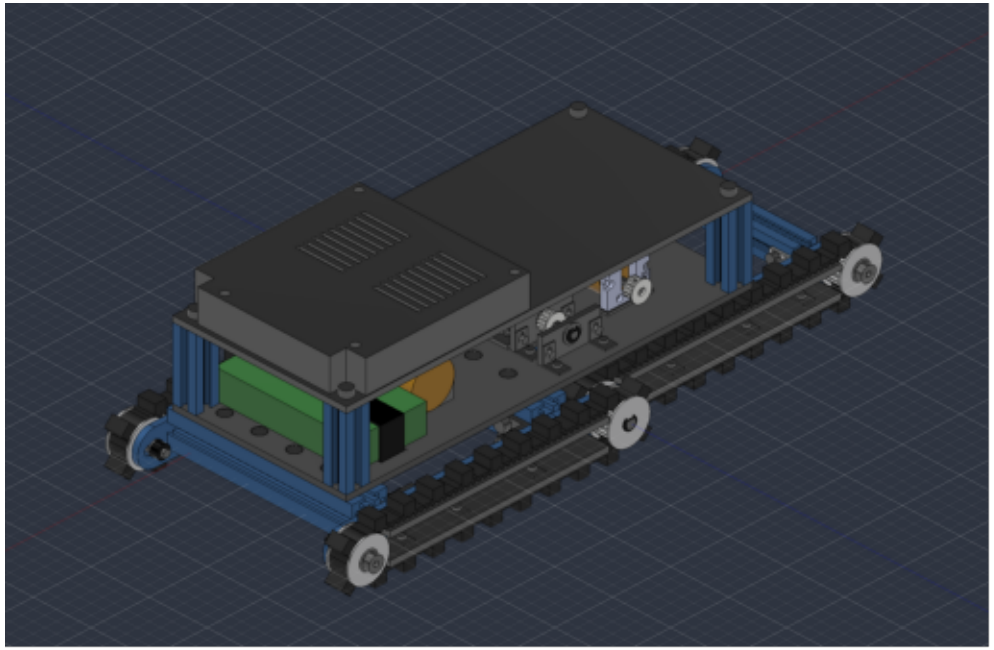
Nhien's Rev B fabrication sheets for the parts the track runs on

Taken unchanged from 'RB4107 Fabrication drawings (Rev B).pdf' (Nhien, 15 Oct 2026). These are the side plates, rails, bars, brackets, hinge supports, shafts, bought-part modifications and the sector gear that the track and wheels mount on, plus his bill of materials, which is the parts list this pack follows.

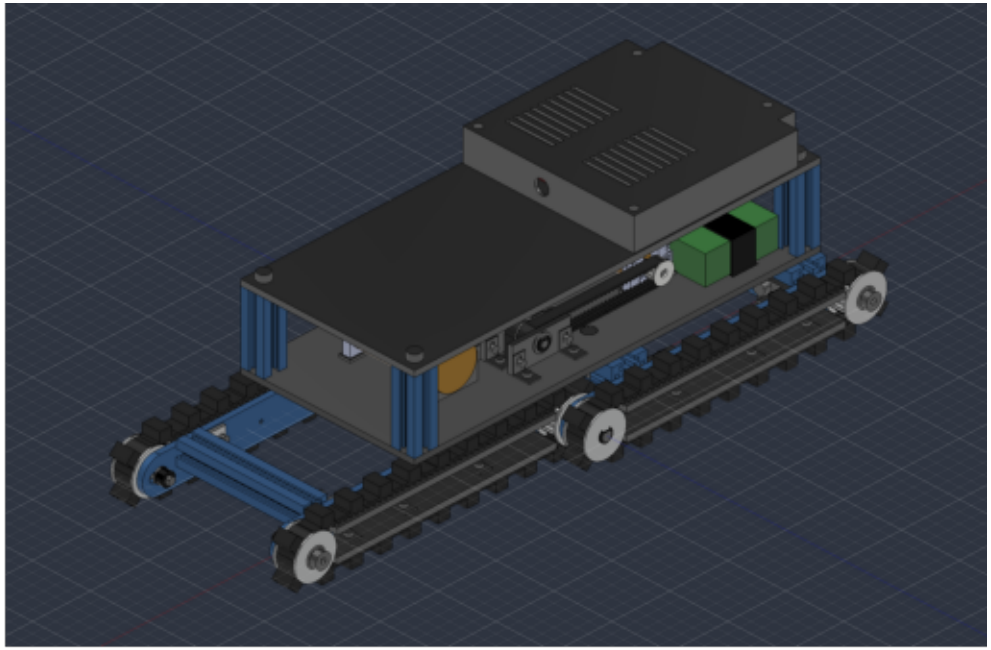
His sheet	Content
2	Assembly views
3	Joint / fastener schedule and assembly order
4	P1 side plate, front segment
5	P2 side plate, rear segment
18	P3 to P6 slide rails (MISUMI UPA plate, drilled)
19	P7 / P8 / P15 profile cut list
12	P18 corner bracket, 3030
13	P19 corner bracket, 2020
11	P17 hinge shaft bearing support
20	P20 / P21 shafts
21	M1 to M8 modifications to bought parts (track pulleys M1, M2)
9	P11 hinge sector gear
14	P14 sector mount angle
23, 24, 25	Bill of materials (also RB4107 BOM Rev B.xlsx)

Not included: deck, tray, drive-train and electronics sheets (P9, P10, P12, P13, P16, P22, P23, M7, M8), which are Nhien's scope and unchanged by the track.

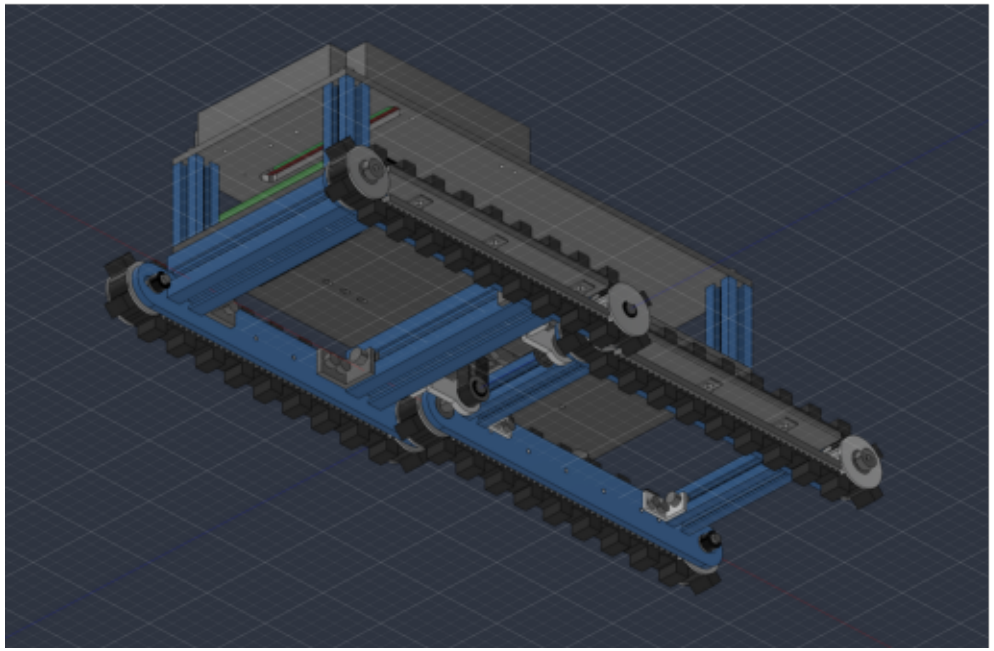
ASSEMBLY VIEWS (reference only)



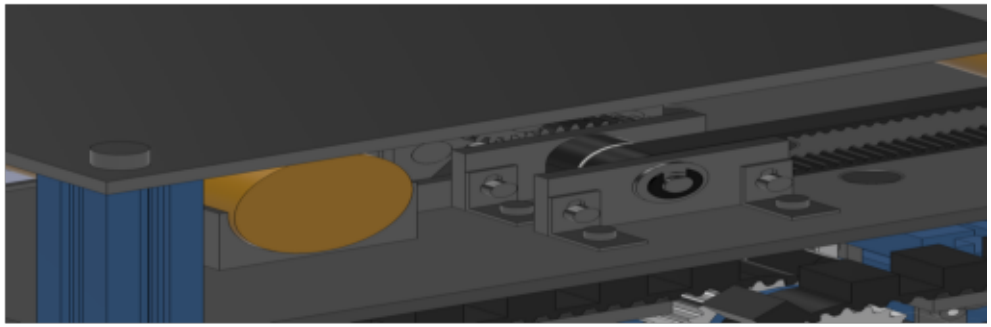
Overall, from the rear, B side



Overall, from the front, A side



From below: track sides, bars, bearings, idler slots



Hinge area

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	ASM Assembly views
MATERIAL	-
QTY	-
SCALE	NTS
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	2 of 25

JOINT / FASTENER SCHEDULE

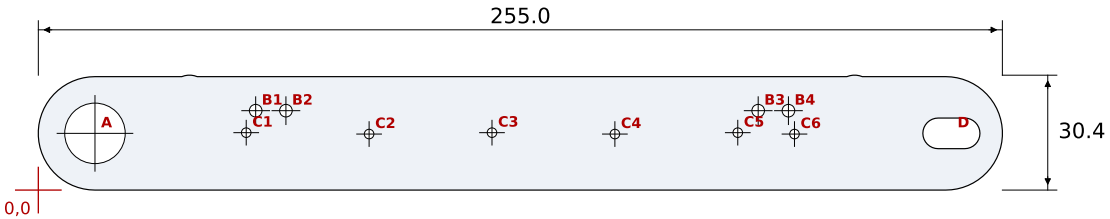
JOINT	FASTENERS
Side plate P2 - 3030 bar P8 (4 corners)	P18 bracket; M6x10 SHCS + slot-8 M6 T-nut; 2x M5x8 SHCS into plate
Side plate P1 - 2020 bar P7 (4 corners)	P19 bracket; M5x8 SHCS + slot-6 M5 T-nut; 2x M4x8 SHCS into plate
Deck P12 - 3030 bars P8 (rear + middle)	8x M5x12 csk ISO 10642 (from top) + slot-8 M5 T-nuts
Deck P12 - posts P15	4x M6x16 csk ISO 10642 from BELOW into tapped post ends
Tray P16 - posts P15	4x M6x16 SHCS ISO 4762 from top into tapped post ends
Hinge bearing support P17 - middle bar P8	2x M5x10 SHCS + slot-8 M5 T-nuts each (2 supports)
Shaft support plate P13 - deck	2x RS 427-991 (M7); M3x5 SHCS to plate (slot) + M3x5 to deck (tapped)
Sector gear P11 - angle P14 - bar P7	2x M3x8 csk into gear; 2x M4x8 csk + slot-6 M4 T-nuts into bar
Rail brackets (24) - side plates / rails	M3x6 SHCS into tapped plate; 3 mm x 5 PT screw into rail
Idlers (4) - side plates	ISO 7379 Ø8x25 M6 shoulder screw, 8x11x1 head washer, 8x11x0.5 shim, spacer 8x10x9, M6 washer + nyloc; 2x 688-2RS
Hinge drive shafts P20 (2)	3x 688-2RS each (rear plate, front plate, P17); E-clip DIN 6799-6 both ends; shim 8x11x1
Intermediate shafts P21 (2)	2x F686ZZ each (flanges out); E-clip DIN 6799-5 + shim 6x8x0.5 both ends
Drive motors (2)	Pololu #1995 bracket on P9 shim; M3x12 SHCS from below + nut; tail cradle P10 (M3 inserts)
Hinge motor 5840	M8 L-bracket; 4x M4x8 csk DIN 7991 into gearbox; 2x M5x8 + washer into deck
Controller	MDD10A on M3x6 nylon standoffs; ESP32 sled; BNO085 on M2.5x5; 5V reg. on M2x5; cover 4x M3x25
Battery	20 mm hook-and-loop strap through the two deck slots

TIGHTENING (A2/8.8 into aluminium): M3 1.0 N·m M4 2.0 N·m M5 4.0 N·m M6 7 N·m. Grub screws: Loctite 243.
ASSEMBLY ORDER
1. Press bearings into side plates, P17 supports and idlers (Loctite 641 on outer ring if loose).
2. Build the rear frame: P2 plates + P8 bars with P18 brackets (drop T-nuts into the bar slots first).
3. Build the front frame: P1 plates + P7 bars with P19 brackets; fit P14 + P11 sector gear.
4. Hinge: slide P20 shafts through rear pulley, rear plate, front pulley, front plate, P17, shim, pulley; E-clips; grubs.
5. Deck P12 onto the bars (M5 csk), posts P15 (M6 from below), tray P16 (M6 from top).
6. Drive: motors + P9 shims + cradles, P13 supports + P21 shafts, belts. Tension Stage-1 (motor slots) then Stage-2 (M7 slots).
7. Idlers on shoulder screws; tension each track by sliding the idler out, then tighten the nyloc nut.
8. Electronics on the tray, wiring through the slot (grommet), cover P22 last.

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	ASM Joint / fastener schedule
MATERIAL	-
QTY	-
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	3 of 25

P1 SIDE PLATE - FRONT SEGMENT

VIEW FROM INSIDE (bar side); hinge end left, nose end right (scale 0.5:1, thickness 5.0)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
A	14.98	15.00	Ø16.0 H7 THRU (688-2RS bearing seat)
B1	57.48	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
B2	65.48	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
B3	190.38	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
B4	198.38	21.00	M4x0.7-6H THRU (tap drill Ø3.3)
C1	54.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C2	87.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C3	119.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C4	152.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C5	184.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C6	199.98	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
D	241.48	15.00	SLOT 8.1 wide x 15.1 long, along X, THRU

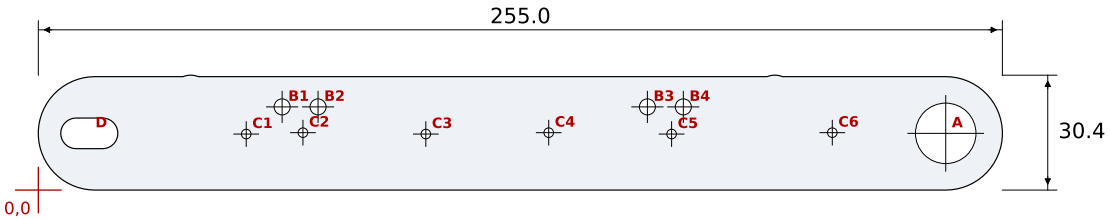
NOTES

1. Laser / waterjet cut from DXF, then drill and tap. Deburr, break edges 0.2-0.3.
2. Opposite-hand plate = mirror image (STEP 'opposite hand').
3. Ø16 H7 at the hinge end: press in a 688-2RS bearing with Loctite 641.
4. Slot at the nose end = idler tension slot for the ISO 7379 Ø8 shoulder screw.
5. M3 tapped holes: RS 427-991 rail brackets (check a sample bracket before tapping).
6. M4 tapped holes: 2020 inside corner brackets P19 (2 per bar end).
7. CAD: STEP/P1*.step DXF/P1*.dxf

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P1 Side plate - FRONT segment
MATERIAL	Al 6061-T6, 5 mm plate
QTY	2 (1 as drawn + 1 opposite hand)
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	4 of 25

P2 SIDE PLATE - REAR SEGMENT

VIEW FROM INSIDE (bar side); rear idler left, hinge end right (scale 0.5:1, thickness 5.0)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
A	239.98	15.00	Ø16.0 H7 THRU (688-2RS bearing seat)
B1	64.48	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
B2	73.98	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
B3	161.08	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
B4	170.58	22.00	M5x0.8-6H THRU (tap drill Ø4.2)
C1	54.98	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C2	69.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C3	102.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C4	134.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
C5	167.48	14.82	M3x0.5-6H THRU (tap drill Ø2.5)
C6	209.98	15.18	M3x0.5-6H THRU (tap drill Ø2.5)
D	13.48	15.00	SLOT 8.1 wide x 15.1 long, along X, THRU

NOTES

1. Laser / waterjet cut from DXF, then drill and tap. Deburr, break edges 0.2-0.3.
2. Opposite-hand plate = mirror image (STEP 'opposite hand').
3. Ø16 H7 at the hinge end: press in a 688-2RS bearing with Loctite 641.
4. Slot at the rear end = idler tension slot for the ISO 7379 Ø8 shoulder screw.
5. M3 tapped holes: RS 427-991 rail brackets. M5 tapped holes: 3030 corner brackets P18.
6. CAD: STEP/P2*.step DXF/P2*.dxf

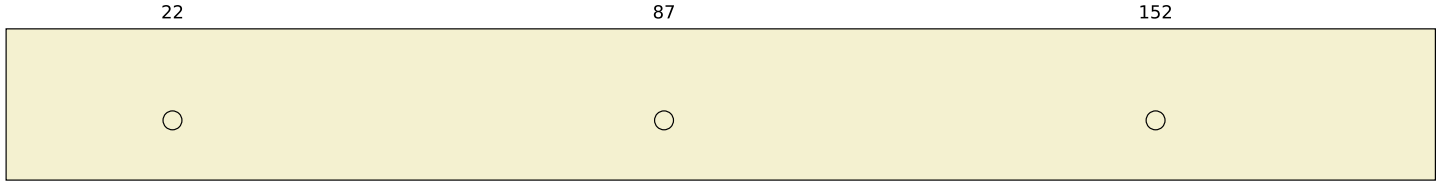
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P2 Side plate - REAR segment
MATERIAL	Al 6061-T6, 5 mm plate
QTY	2 (1 as drawn + 1 opposite hand)
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	5 of 25

P3-P6 SLIDE RAILS = MISUMI UPA-189-20-5 UHMW-PE plate (bought, drill only)

P3

Slide rail - FRONT, UPPER

QTY 1 + 1 opp. hand

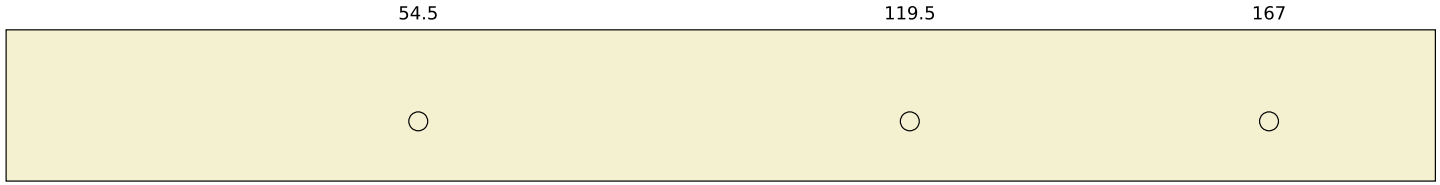


189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

P4

Slide rail - FRONT, LOWER

QTY 1 + 1 opp. hand

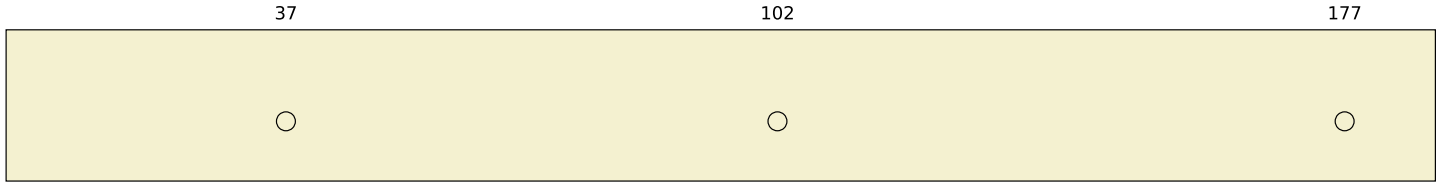


189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

P5

Slide rail - REAR, UPPER

QTY 1 + 1 opp. hand

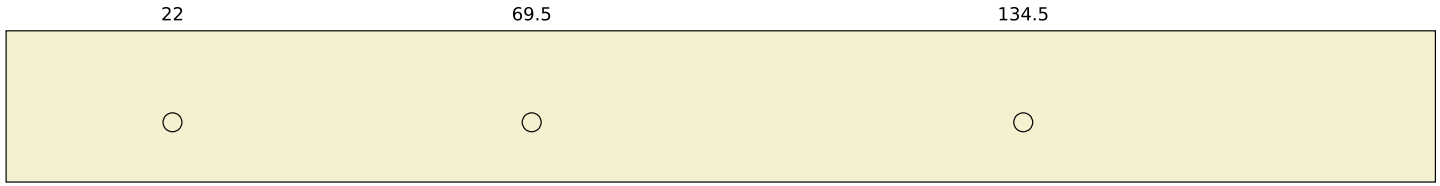


189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

P6

Slide rail - REAR, LOWER

QTY 1 + 1 opp. hand



189 x 20 x 5; 3x Ø2.5 x 4 deep (blind), hole centres 7.9 from the INNER (side-plate) edge - drawn as the lower edge

NOTES

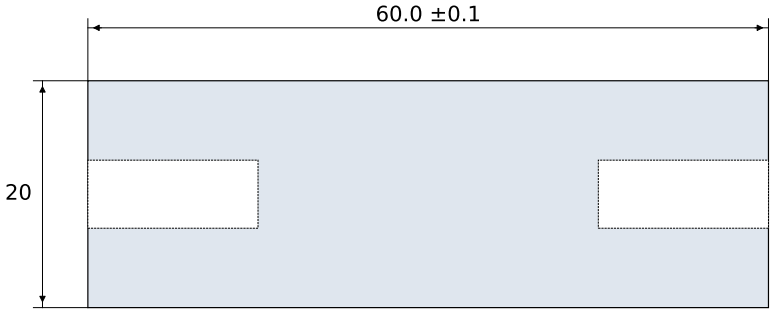
1. Order MISUMI UPA-189-20-5 (UHMW-PE plate, milky white, 189 x 20 x 5, cut to size) x 8. Only the 3 pilot holes are drilled.
   The strip is 5 mm wider than the 15 mm belt; the extra 5 mm overhangs on the OUTER side (does not touch the belt).
   Pilot holes must not break through the wear face.
2. 3 mm thread-forming screws for plastics (PT type) x 5, through the RS 427-991 bracket legs.

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P3-P6 Slide rails (MISUMI UPA-189-20-5)
MATERIAL	UHMW-PE 189 x 20 x 5
QTY	1 + 1 opp. each (8)
SCALE	1:1
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	18 of 25

P7 / P8 / P15 ALUMINIUM T-SLOT PROFILES - CUT LIST

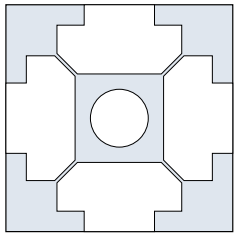
Nothing screws into the SIDE of a profile: side joints use T-nuts in the slots. Only P15 ends are tapped.

PART	DESCRIPTION	PROFILE	LENGTH	QTY	MACHINING
P7	Cross bar - front segment (hinge-side + nose)	2020 B-type, slot 6, bore Ø5.1	110.0 ±0.2	2	Cut square, deburr. No machining.
P8	Cross bar - rear + middle	3030 B-type, slot 8, bore Ø6.65 (80/20 30-3030 type)	160.0 ±0.2	2	Cut square, deburr. No machining.
P15	Corner post (deck to tray)	2020 B-type, slot 6, bore Ø5.1	60.0 ±0.1	4	Tap M6x1 15 deep BOTH ends (centre bore).

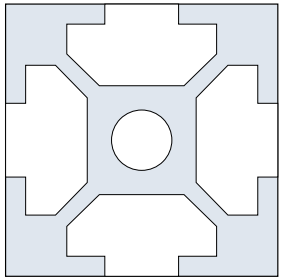


P15 corner post (1.5:1)

M6x1-6H x 15 deep, both ends, in the Ø5.1 centre bore



2020 section (1.5:1)



3030 section (1.2:1)

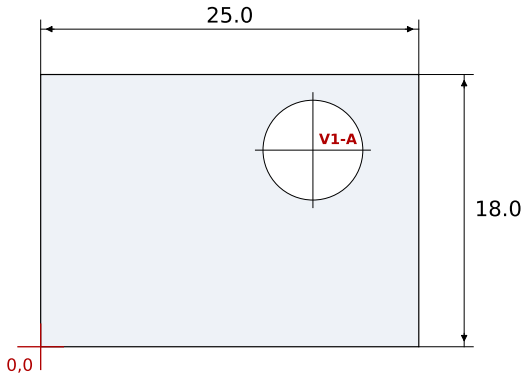
NOTES

1. Profiles: 6063-T5 / 6061 extrusion, B-type slots. Order cut-to-length from the profile supplier if possible.
2. Ends square to the axis within 0.1 so plates and brackets sit flat.
3. P8 ends are NOT tapped (Rev A tapped M5 - removed). They are fixed with P18 corner brackets + M6 T-nuts.
4. P7 ends are NOT tapped any more: fixed with P19 corner brackets + M5 T-nuts.
5. T-nuts to drop in BEFORE closing the frame: P8 rear: 4x M5 (deck, top slot) + 2x M6 (brackets, inner side slot).
   P8 middle: 4x M5 (deck, top) + 2x M6 (brackets, rear side) + 4x M5 (hinge bearing supports P17, front side).
   P7 hinge-side: 2x M4 (sector angle, top) + 2x M5 (brackets). P7 nose: 2x M5 (brackets).
6. Use drop-in (roll-in) T-nuts so they can be added after assembly if one is forgotten.

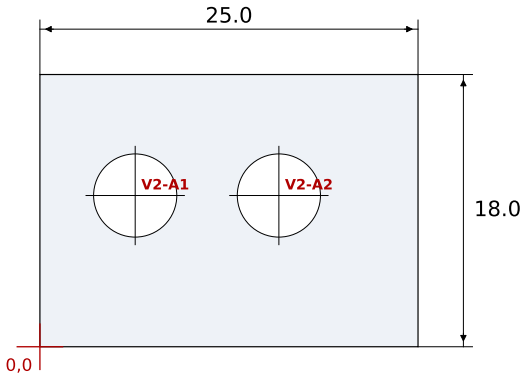
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P7/P8/P15 Profile cut list
MATERIAL	Al T-slot extrusion
QTY	2 / 2 / 4
SCALE	as shown
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	19 of 25

P18 INSIDE CORNER BRACKET - 3030 BARS

LEG ON BAR SIDE (T-nut) (scale 2:1)



LEG ON SIDE PLATE (scale 2:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A	18.00	13.00	Ø6.6 THRU (M6 clearance)
V2-A1	6.30	10.00	Ø5.5 THRU (M5 clearance)
V2-A2	15.80	10.00	Ø5.5 THRU (M5 clearance)

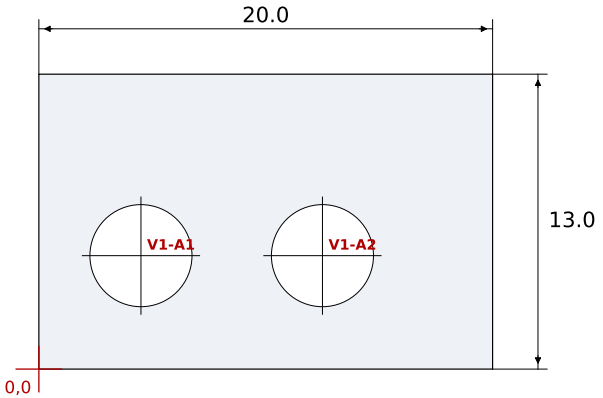
NOTES

1. Cut 18 long from 25x25x4 aluminium angle, drill only.
2. Bar leg: Ø6.6 for M6x10 + slot-8 M6 T-nut. Plate leg: 2x Ø5.5 for M5x8 into the tapped side plate.
3. CAD: STEP/P18*.step

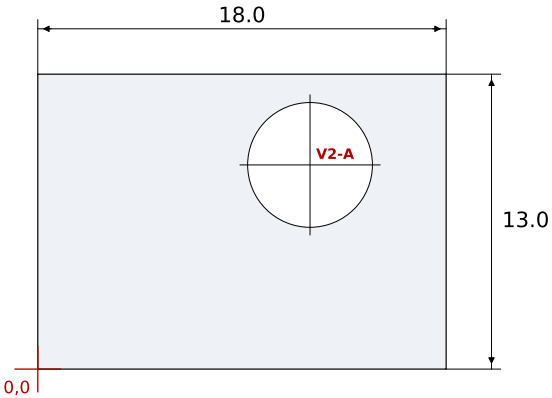
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P18 Inside corner bracket - 3030 bars
MATERIAL	Al angle 25x25x4 (6061/6063)
QTY	4
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	12 of 25

P19 INSIDE CORNER BRACKET - 2020 BARS

LEG ON SIDE PLATE (scale 3:1)



LEG ON BAR SIDE (T-nut) (scale 3:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A1	4.50	5.00	Ø4.5 THRU (M4 clearance)
V1-A2	12.50	5.00	Ø4.5 THRU (M4 clearance)
V2-A	12.00	9.00	Ø5.5 THRU (M5 clearance)

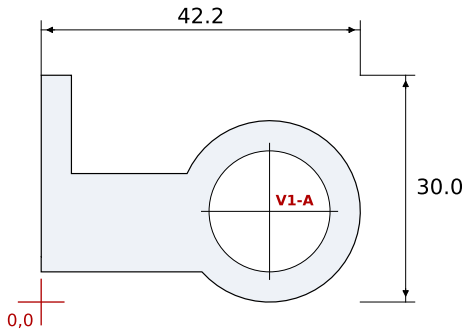
NOTES

1. Cut 13 long from 20x20x3 aluminium angle; trim the plate leg to 18. Drill only.
2. Plate leg: 2x Ø4.5 for M4x8. Bar leg: Ø5.5 for M5x8 + slot-6 M5 T-nut.
3. CAD: STEP/P19*.step

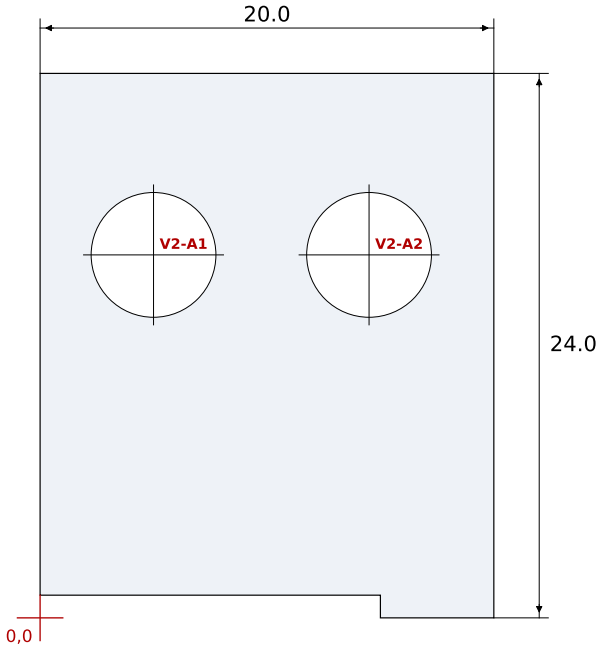
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P19 Inside corner bracket - 2020 bars
MATERIAL	Al angle 20x20x3 (6061/6063)
QTY	4
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	13 of 25

P17 HINGE SHAFT BEARING SUPPORT A (B = OPPOSITE HAND)

PLATE FACE (scale 1:1)



FOOT FACE (against 3030 bar side) (scale 3:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A	30.20	11.99	Ø16.0 H7 THRU (688-2RS bearing seat)
V2-A1	5.00	16.00	Ø5.5 THRU (M5 clearance)
V2-A2	14.50	16.00	Ø5.5 THRU (M5 clearance)

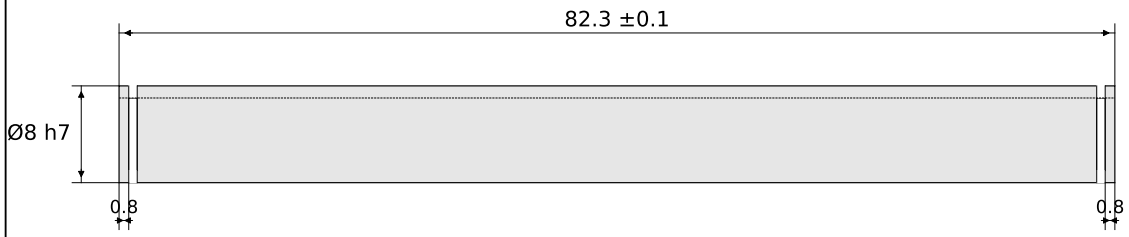
NOTES

1. One piece: 5 mm plate + 4 mm foot (machined from 25 x 45 bar, or 3D print in PETG-CF, 8 mm plate).
2. Ø16 H7 bearing seat for 688-2RS. Foot: 2x Ø5.5 for M5x10 into slot-8 T-nuts on the middle bar.
3. CAD: STEP/P17*.step DXF/P17*.dxf

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P17 Hinge shaft bearing support A (B = opposite hand)
MATERIAL	Al 6061-T6 (or PETG-CF print)
QTY	1 + 1 opposite hand
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	11 of 25

P20 / P21 SHAFTS (D-shaft stock, cut + groove)

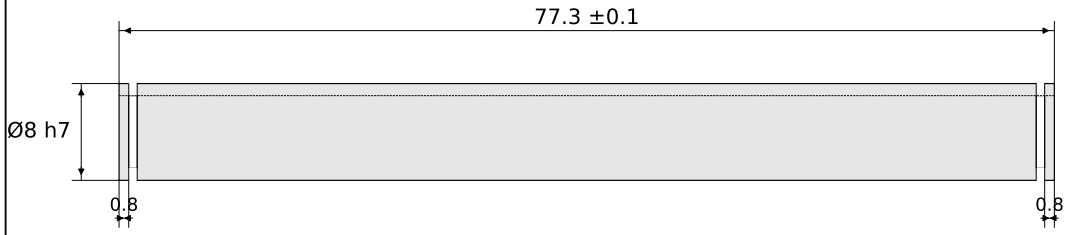
Start from ground D-shaft stock (Ø8 flat 7, Ø6 flat 5.5). Cut to length, chamfer 0.3x45°, turn the two E-clip grooves.



P20-A Hinge drive shaft A (QTY 1)

D-flat 7 across, FULL LENGTH (dashed)

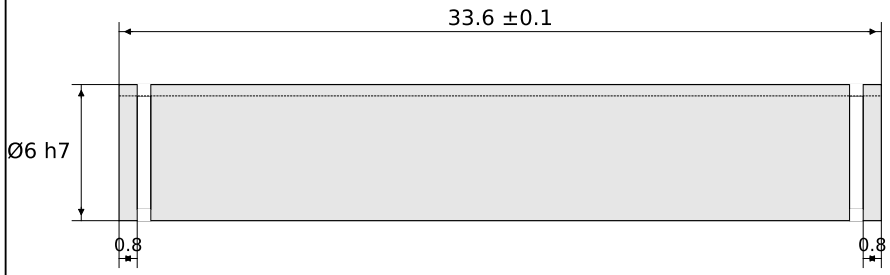
2x groove Ø6 x 0.7 wide for E-clip DIN 6799-6



P20-B Hinge drive shaft B (QTY 1)

D-flat 7 across, FULL LENGTH (dashed)

2x groove Ø6 x 0.7 wide for E-clip DIN 6799-6



P21 Intermediate shaft (QTY 2)

D-flat 5.5 across, FULL LENGTH (dashed)

2x groove Ø5 x 0.6 wide for E-clip DIN 6799-5

NOTES

1. Material: steel D-shaft (1045 / 303 stainless, ground h7). Hardness not required.
2. Grooves: P20 Ø6.0 +0/-0.06 x 0.7 (DIN 6799-6); P21 Ø5.0 +0/-0.06 x 0.6 (DIN 6799-5). Sharp groove corners.
3. Bearings (688-2RS / F686ZZ) slide over the D; fix inner rings with Loctite 641 at assembly.
4. Stack on P20 (from one end): rear drive pulley - rear-plate bearing - front drive pulley - front-plate bearing - support bearing (P17) - shim 8x11x1 - stage-2 pulley. E-clips at both ends trap the stack.
5. Front drive pulleys also get an M4x8 cup-point grub screw onto the flat (Loctite 243).

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P20/P21 Shafts
MATERIAL	Steel D-shaft Ø8 / Ø6
QTY	1+1 / 2
SCALE	1.6:1, 3:1
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	20 of 25

M1-M8 MODIFICATIONS TO BOUGHT / EXISTING PARTS

ID	PART	QTY	MODIFICATION
M1	Track drive pulley HTD 20T BF (rear x2, front x2)	4	Bore out pilot to Ø8 D (flat 7), or order D-bore 8. Keep both flanges (Ø36 x 1.5).
			FRONT pair only: radial M4 tapped hole at mid-width (9.5 from a flange face), onto the flat (M4x8 grub).
M2	Track idler pulley HTD 20T BF (rear x2, nose x2)	4	Bore Ø16 H7 through for 2x 688-2RS (press fit) + spacer sleeve 8x10x9 between them.
M3	Stage-2 pulley on hinge shaft (HTD 5M)	2	Bore: Ø8 D (flat 7).
M4	Intermediate pulleys, stage 1 + stage 2	4	Bore: Ø6 D (flat 5.5). M3 grub screw optional.
M5	Motor pulley, stage 1 (HTD 3M)	2	Radial M3 tapped hole mid-width onto the motor D-flat (M3x3 grub, max length 3).
M6	Hinge pinion m1 20T, 5 mm face	1	Bore: Ø8 D (flat 7) to suit the 5840 shaft. Radial M3 tapped hole (M3x4 grub).
M7	RS PRO 427-991 bracket (shaft supports)	8	Upright-leg hole elongated upward to a 3.4 x 6.4 slot (belt tension). Sheet M7.
M8	5840 L-bracket (motor kit)	1	4x Ø4.5 csk Ø8.4 x 90° on 40 x 28 (gearbox datasheet). Sheet M8.

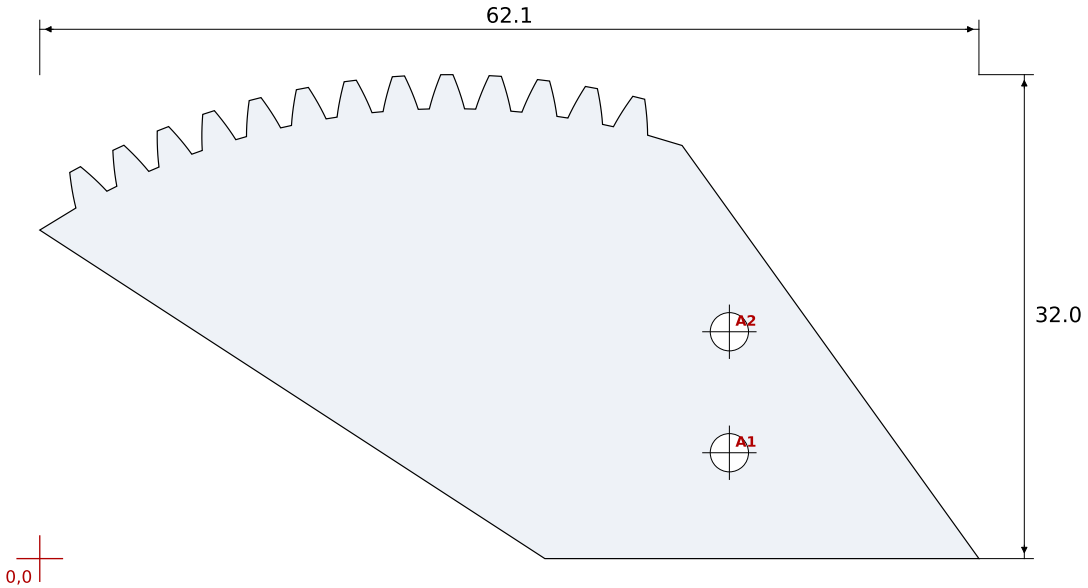
NOTES

1. Grub screws: ISO 4029 cup point, Loctite 243. Radial holes at mid-width of the hub/teeth, between teeth if possible.
2. If a sprocket / pulley is 3D printed, print the D-bore and use heat-set inserts for the grub screws.
3. Relief faces: only the hub boss may touch a bearing inner ring; nothing rotating may rub on a plate.
4. STEP files of each modified part are in STEP/M*.step for reference.
5. M1/M2 = standard HTD 5M 20T BF pulley for 15 mm belt: OD 30.69, PD 31.83, toothed 16, 2 flanges Ø36 x 1.5, total 19.
   M2 bore Ø16 is above the catalogue max (14) - leaves a 5.3 wall under the teeth; ask the supplier/lathe to bore it.

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	M1-M8 Modified bought parts
MATERIAL	various
QTY	see table
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	21 of 25

P11 HINGE SECTOR GEAR M1 R51 SEGMENT

VIEW FROM +Z (motor side). Origin of CAD = hinge axle   (scale 2:1, thickness 5.0)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
A1	45.60	7.00	M3x0.5-6H THRU (tap drill Ø2.5)
A2	45.60	15.00	M3x0.5-6H THRU (tap drill Ø2.5)

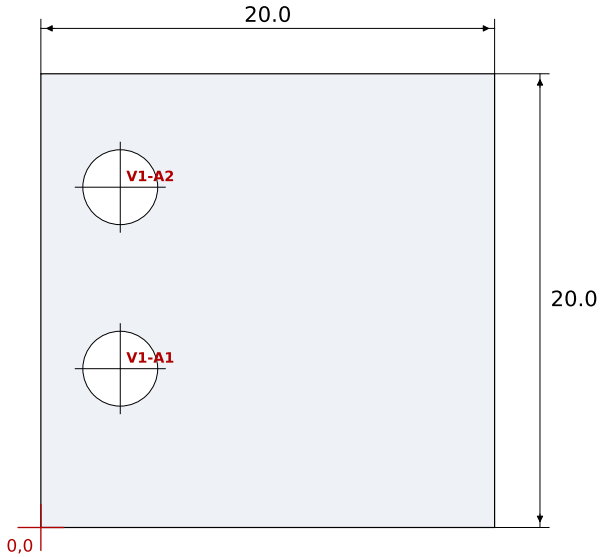
NOTES

1. Laser / wire cut from DXF (tooth flanks in the DXF are true involute, m1, 20° PA, pitch R51).
2. Rev B: old Ø4.5 holes removed. 2x M3 tapped holes for the P14 angle (M3x8 csk from the angle side).
3. Screw tips must end flush with the outer face (hinge bracket plate is 1 mm away).
4. CAD: STEP/P11*.step   DXF/P11*.dxf

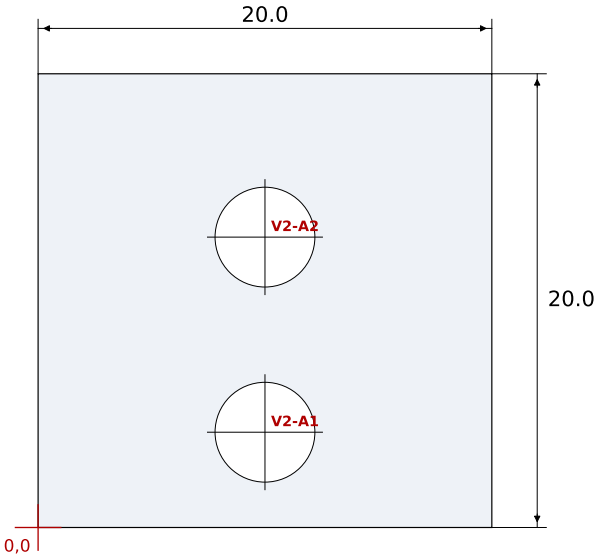
PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P11 Hinge sector gear m1 R51 segment
MATERIAL	Steel S275 / mild, 5 mm
QTY	1
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	9 of 25

P14 SECTOR MOUNT ANGLE

VERTICAL LEG (against sector gear) (scale 3:1)



BASE LEG (on 2020 bar) (scale 3:1)



HOLE / FEATURE TABLE (X, Y from the red 0,0 datum)

ID	X	Y	DESCRIPTION
V1-A1	3.50	7.00	M4x0.7-6H THRU (tap drill Ø3.3), CSK Ø6.5x90° THIS FACE
V1-A2	3.50	15.00	M4x0.7-6H THRU (tap drill Ø3.3), CSK Ø6.5x90° THIS FACE
V2-A1	10.00	4.20	Ø4.4 THRU (M4 clearance), CSK Ø4.4x90° THIS FACE
V2-A2	10.00	12.80	Ø4.4 THRU (M4 clearance), CSK Ø4.4x90° THIS FACE

NOTES

1. Cut 20 long from 20x20x3 angle. Drill + countersink.
2. Vertical leg: 2x M3 csk from the INSIDE of the angle (M3x8 into the tapped sector gear).
3. Base leg: 2x M4 csk on top, on the bar slot centre line (M4x8 + slot-6 T-nuts).
4. CAD: STEP/P14*.step

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	P14 Sector mount angle
MATERIAL	Al angle 20x20x3
QTY	1
SCALE	see view
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	14 of 25

BILL OF MATERIALS (1/3) - also in RB4107 BOM.xlsx

ID	DESCRIPTION	SPEC / STANDARD	QTY	SOURCE
== 1 Fabricated parts (make) ==				
P1	Side plate - front segment (1 + 1 opposite hand)	Al 6061-T6 5 mm, laser + drill/tap	2	make - drawings
P2	Side plate - rear segment (1 + 1 opposite hand)	Al 6061-T6 5 mm, laser + drill/tap	2	make - drawings
P3-P6	Slide rails / wear strips: MISUMI UPA-189-20-5, drill 3 pilot holes	UHMW-PE 189 x 20 x 5	8	make - drawings
P7	Cross bar, front segment (hinge-side + nose)	2020 profile x 110	2	make - drawings
P8	Cross bar, rear + middle	3030 profile x 160	2	make - drawings
P9	Motor shim plate, slotted	Al 2 mm, laser	2	make - drawings
P10	Drive-motor tail cradle	PETG print	2	make - drawings
P11	Hinge sector gear, m1 R51 segment	Steel 5 mm, laser/wire	1	make - drawings
P12	Deck plate	Al 6061-T6 5 mm, laser + drill/tap	1	make - drawings
P13	Intermediate shaft support plate (A1, A2, B1, B2)	Al 6061-T6 4 mm	4	make - drawings
P14	Sector mount angle	Al angle 20x20x3 x 20	1	make - drawings
P15	Corner post, M6 tapped both ends	2020 profile x 60	4	make - drawings
P16	Tray plate (controller deck)	Al 6061-T6 5 mm, laser + drill/tap	1	make - drawings
P17	Hinge shaft bearing support (A + B opposite hand)	Al 6061 or PETG-CF	2	make - drawings
P18	Inside corner bracket, 3030	Al angle 25x25x4 x 18	4	make - drawings
P19	Inside corner bracket, 2020	Al angle 20x20x3 x 13	4	make - drawings
P20	Hinge drive shaft A 82.3 / B 77.3	Steel D-shaft Ø8 (flat 7)	2	make - drawings
P21	Intermediate shaft 33.6	Steel D-shaft Ø6 (flat 5.5)	2	make - drawings
P22	Controller cover, one-piece, vented	PETG print	1	make - drawings
P23	ESP32 sled	PETG print	1	make - drawings
== 2 Bought parts to modify ==				
M1	Track drive pulley: D-bore 8 (+ M4 grub, front pair)	HTD5M 20T BF, 15 mm belt	4	buy + modify
M2	Track idler: Ø16 H7 bore for 2x 688-2RS	HTD5M 20T BF, 15 mm belt	4	buy + modify
M3	Stage-2 pulley on hinge shaft: D-bore 8	HTD 5M, 9 mm belt	2	buy + modify
M4	Intermediate pulleys (stage 1 + 2): D-bore 6	HTD 3M / 5M	4	buy + modify
M5	Motor pulley stage 1: radial M3 grub hole	HTD 3M, 6 mm D bore	2	buy + modify
M6	Hinge pinion: D-bore 8 + M3 grub hole	Spur gear m1 20T, 20° PA, 5 face	1	buy + modify
M7	RS PRO 427-991 bracket: leg hole -> slot (shaft supports)	see sheet M7	8	buy + modify
M8	5840 L-bracket: 4x Ø4.5 csk on 40 x 28	see sheet M8	1	buy + modify
== 3 Bought mechanical parts ==				
B1	Gearmotor 37D, 12 V, with encoder (drive A + B)	Pololu #2828	2	Pololu
B2	Machined bracket for 37D gearmotor	Pololu #1995	2	Pololu
B3	Worm gear motor 5840-31ZY 12 V + L-bracket kit (hinge)	5840-31ZY	1	AliExpress / RobotShop
B4	Ball bearing 688-2RS 8 x 16 x 5	688-2RS	14	RS / Misumi
B5	Flanged ball bearing F686ZZ 6 x 13 x 5 (flange 15 x 1.1)	F686ZZ	4	RS / Misumi
B6	Shoulder screw Ø8 x 25, M6 thread (idler stub axle)	ISO 7379 8x25	4	Misumi / RS
B7	Angle bracket 15 x 14.6 x 10, 1.2 mm steel (rails + shaft supports)	RS PRO 427-991	32	RS

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	BOM Bill of materials
MATERIAL	-
QTY	-
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
DATE / REV	2026-10-15 / Rev B
SHEET	23 of 25

BILL OF MATERIALS (2/3) - also in RB4107 BOM.xlsx

ID	DESCRIPTION	SPEC / STANDARD	QTY	SOURCE
B8	Stage-1 timing belt (motor -> intermediate)	HTD 3M, 9 wide, length per Rev A	2	per Rev A spec
B9	Stage-2 timing belt (intermediate -> hinge shaft)	HTD 5M, 9 wide, length per Rev A	2	per Rev A spec
B10	Track belt HTD 550-5M-15 (110 T) + 22 cleats every 5 teeth (25 mm), cleat 10 x	e.g. SKF PHG 550-5M-15, RS 058-6535	4	RS + cleats by belt sh
B11	Al 2020 B-type profile, slot 6, bore Ø5.1 (P7 + P15) - stock	2020, 1 m	1	Misumi / local
B12	Al 3030 B-type profile, slot 8 (80/20 30-3030 type) (P8) - stock	3030, 0.5 m	1	8020 / local
B13	Steel D-shaft Ø8, flat 7 (P20) - stock 200 mm	Ø8 h7 D-shaft	1	Misumi / local
B14	Steel D-shaft Ø6, flat 5.5 (P21) - stock 100 mm	Ø6 h7 D-shaft	1	Misumi / local
B15	Al angle 25x25x4 (P18) - 100 mm	6063	1	local
B16	Al angle 20x20x3 (P14, P19) - 100 mm	6063	1	local
B17	Battery strap 20 mm hook-and-loop with buckle, ~230 mm	Velcro	1	local
B18	Foam pad 10 x 10 x 8, adhesive-backed (hinge up-stops)	EVA / neoprene	2	local
B19	Cable gland PG7 (actuator / hinge leads through cover)	PG7, 3-6.5 mm cable	1	RS
B20	Grommet edge strip for tray wire slot, 0.4 m	1-2 mm panel	1	RS
== 4 Fasteners ==				
F1	Socket head cap screw M3 x 5	ISO 4762 A2	16	RS / Misumi / local
F2	Socket head cap screw M3 x 6 (rail brackets to plates)	ISO 4762 A2	24	RS / Misumi / local
F3	Thread-forming screw for plastics 3 x 5, pan head (rails)	PT / Delta-PT	24	RS / Misumi / local
F4	Countersunk screw M3 x 8	ISO 10642	2	RS / Misumi / local
F5	Countersunk screw M4 x 8	ISO 10642	2	RS / Misumi / local
F6	Countersunk screw M4 x 8 (bracket to 5840 gearbox)	DIN 7991	4	RS / Misumi / local
F7	Socket head cap screw M4 x 8	ISO 4762	8	RS / Misumi / local
F8	Socket head cap screw M5 x 8	ISO 4762	12	RS / Misumi / local
F9	Socket head cap screw M5 x 10	ISO 4762	4	RS / Misumi / local
F10	Countersunk screw M5 x 12	ISO 10642	8	RS / Misumi / local
F11	Socket head cap screw M6 x 10	ISO 4762	4	RS / Misumi / local
F12	Socket head cap screw M6 x 16	ISO 4762	4	RS / Misumi / local
F13	Countersunk screw M6 x 16	ISO 10642	4	RS / Misumi / local
F14	Grub screw M3 x 3, cup point	ISO 4029	2	RS / Misumi / local
F15	Grub screw M3 x 4, cup point	ISO 4029	1	RS / Misumi / local
F16	Grub screw M4 x 8, cup point	ISO 4029	2	RS / Misumi / local
F17	T-nut M4, slot 6 (2020), drop-in	B-type slot 6	2	RS / Misumi / local
F18	T-nut M5, slot 6 (2020), drop-in	B-type slot 6	4	RS / Misumi / local
F19	T-nut M5, slot 8 (3030), drop-in	B-type slot 8	12	RS / Misumi / local
F20	T-nut M6, slot 8 (3030), drop-in	B-type slot 8	4	RS / Misumi / local
F21	E-clip for Ø6 shaft (groove Ø5)	DIN 6799-5	4	RS / Misumi / local
F22	E-clip for Ø8 shaft (groove Ø6)	DIN 6799-6	4	RS / Misumi / local
F23	Shim washer 6 x 8 x 0.5	DIN 988	4	RS / Misumi / local
F24	Shim washer 8 x 11 x 1	DIN 988	6	RS / Misumi / local

PROJECT	RB4107 Safe Evacuation Robot - 1:5 prototype
PART	BOM Bill of materials
MATERIAL	-
QTY	-
SCALE	-
UNITS / TOL.	mm / ISO 2768-m unless stated
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BILL OF MATERIALS (3/3) - also in RB4107 BOM.xlsx

ID	DESCRIPTION	SPEC / STANDARD	QTY	SOURCE
F24a	Shim washer 8 x 11 x 0.5	DIN 988	4	RS / Misumi / local
F24b	Bearing spacer sleeve 8 x 10 x 9 (Al tube, cut)	Al tube 10 x 1	4	RS / Misumi / local
F25	Washer M6	DIN 125 A	4	RS / Misumi / local
F26	Nyloc nut M6	DIN 985	4	RS / Misumi / local
F27	Nylon standoff M3 x 6, male-female (MDD10A)	PA66	8	RS / Misumi / local
F28	Nylon standoff M2.5 x 5, male-female (BNO085)	PA66	4	RS / Misumi / local
F29	Nylon standoff M2 x 5, male-female (5V regulator)	PA66	2	RS / Misumi / local
F30	Socket head cap screw M3 x 25 (controller cover) [not modelled]	ISO 4762	4	RS / local
F31	Nylon screw M3 x 6 (MDD10A to standoffs) [not modelled]	PA66	8	RS / local
F32	Nylon screw M2.5 x 5 (BNO085) + M2 x 4 (regulator) [not modelled]	PA66	6	RS / local
F33	Socket head cap screw M3 x 8 (ESP32 sled to tray) [not modelled]	ISO 4762	2	RS / local
F34	Socket head cap screw M3 x 12 + nut (Pololu bracket base, from below deck) [n	ISO 4762 / ISO 4032	6	RS / local
F35	Heat-set insert M3 x 5 + SHCS M3 x 10 (tail cradles) [not modelled]	brass insert	4	RS / local
F36	Socket head cap screw M5 x 8 + washer (5840 bracket foot) [not modelled]	ISO 4762 + DIN 125	2	RS / local
== 5 Electronics ==				
E1	Dual motor driver 10 A (drive A+B / hinge + actuator)	Cytron MDD10A	2	Cytron
E2	Microcontroller board	ESP32-DevKitC (WROOM-32)	1	Espressif / local
E3	9-axis IMU breakout	BNO085 (Adafruit 4754 size)	1	Adafruit / local
E4	5 V 5 A step-down regulator	Pololu D24V50F5	1	Pololu
E5	Battery 3S LiPo 3000 mAh 'shorty'	11.1 V	1	local
E6	Wiring, connectors, fuse 15 A, power switch	per wiring diagram	1	local
== 6 Consumables ==				
C1	Thread-locker, medium strength (grub screws)	Loctite 243	1	local
C2	Retaining compound (bearings on D-shafts / seats)	Loctite 641	1	local
C3	Cable ties 100 mm	-	20	local

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